

电子信息工程中的数学模型和方法

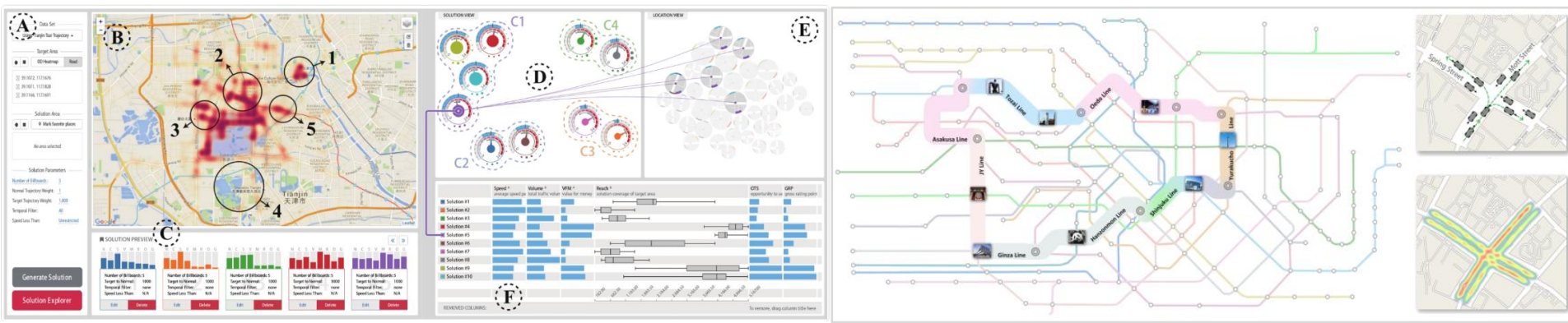
可视化建模（上）

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zhx@cad.zju.edu.cn

<http://www.cad.zju.edu.cn/home/zhx>





数据可视化与可视分析

什么是数据可视化？为什么我们要用数据可视化？

<https://github.com/hongxin/vizmodeling>

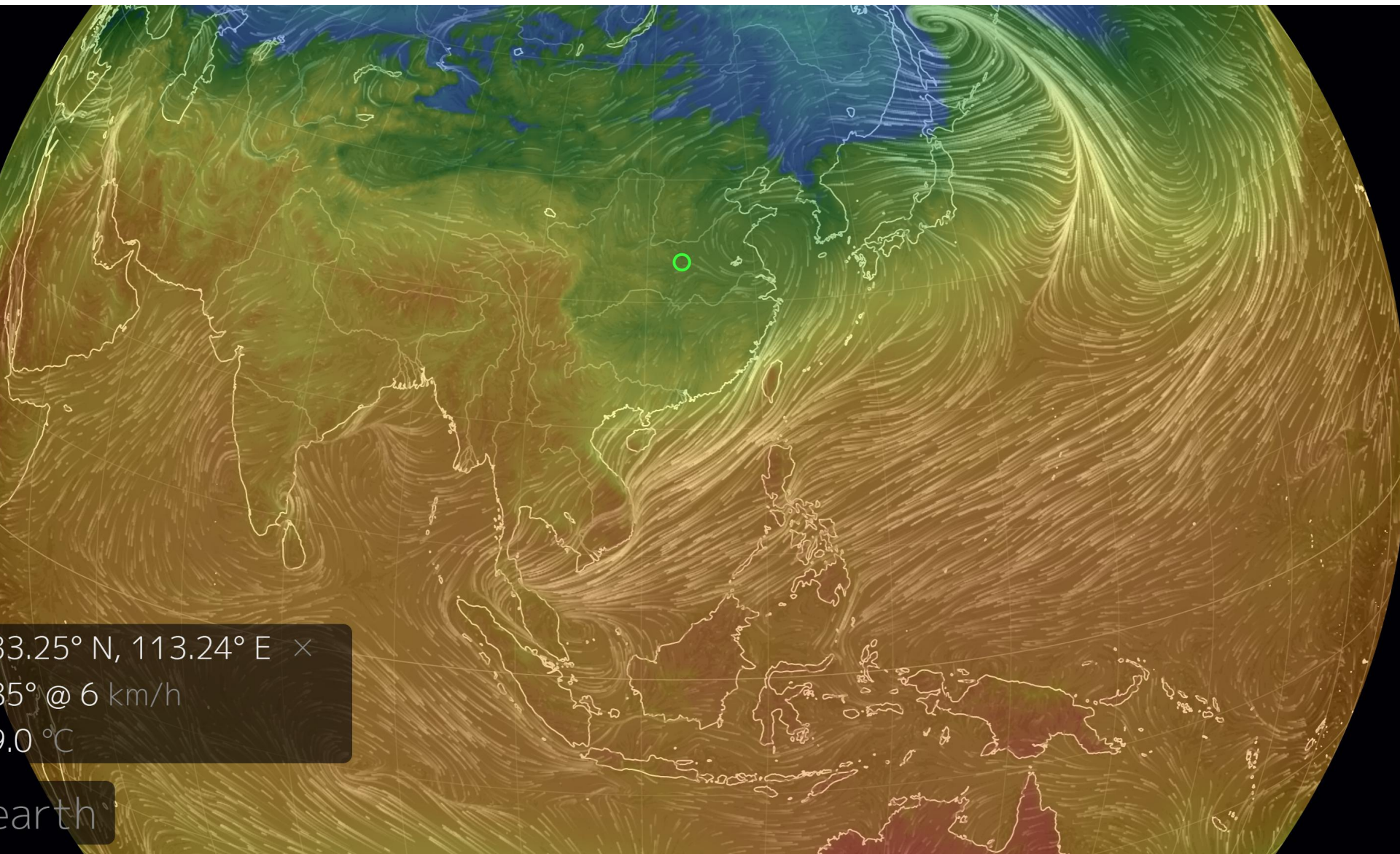


例子 – 飞行模式可视化



例子 – 全球气象可视化

<https://earth.nullschool.net>





例子 – 《本草纲目》可视化

药症方关联的中医药古籍 交互可视分析方法

吴泓嘉, 张弛, 张宏鑫, 陈为, 夏佳志

ChinaVis 2023

ID 1239

本文提出了一种药症方关联的
中医药古籍交互可视分析方法

吴泓嘉, 张弛, 张宏鑫, 陈为, 夏佳志. 药症方关联的中医药古籍交互可视分析方法[J].
计算机辅助设计与图形学学报. DOI: 10.3724/SP.J.1089.2023-00623

什么是数据可视化？

- 创建并研究数据的**视觉表达** (Visual Representation)
 - 输入：数据 (data)
 - 输出：视觉形式 (visual form)
 - 目标：深入理解 (insight)



数据



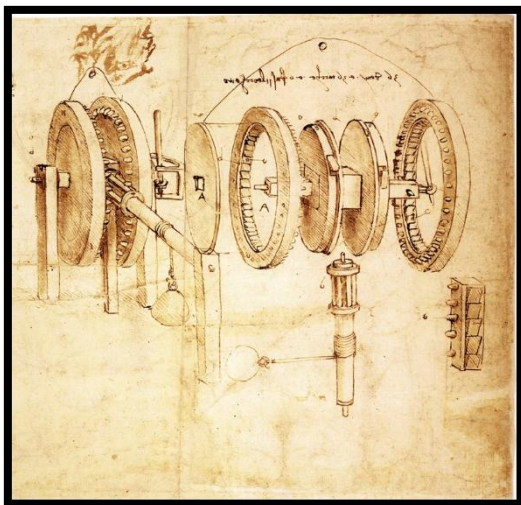
视觉形式



深入理解

数据可视化的主要任务

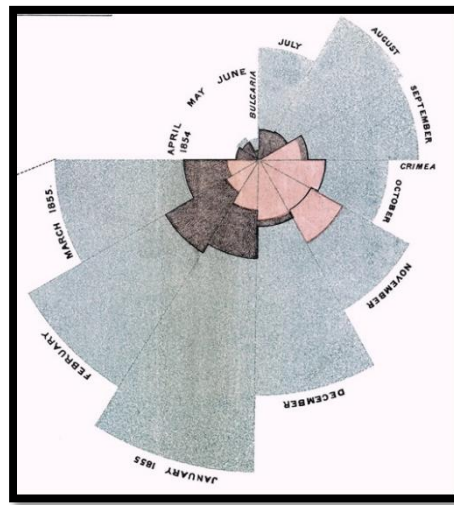
- 表示数据 - Represent
- 分析数据 - Analyze
- 交流数据 - Communicate



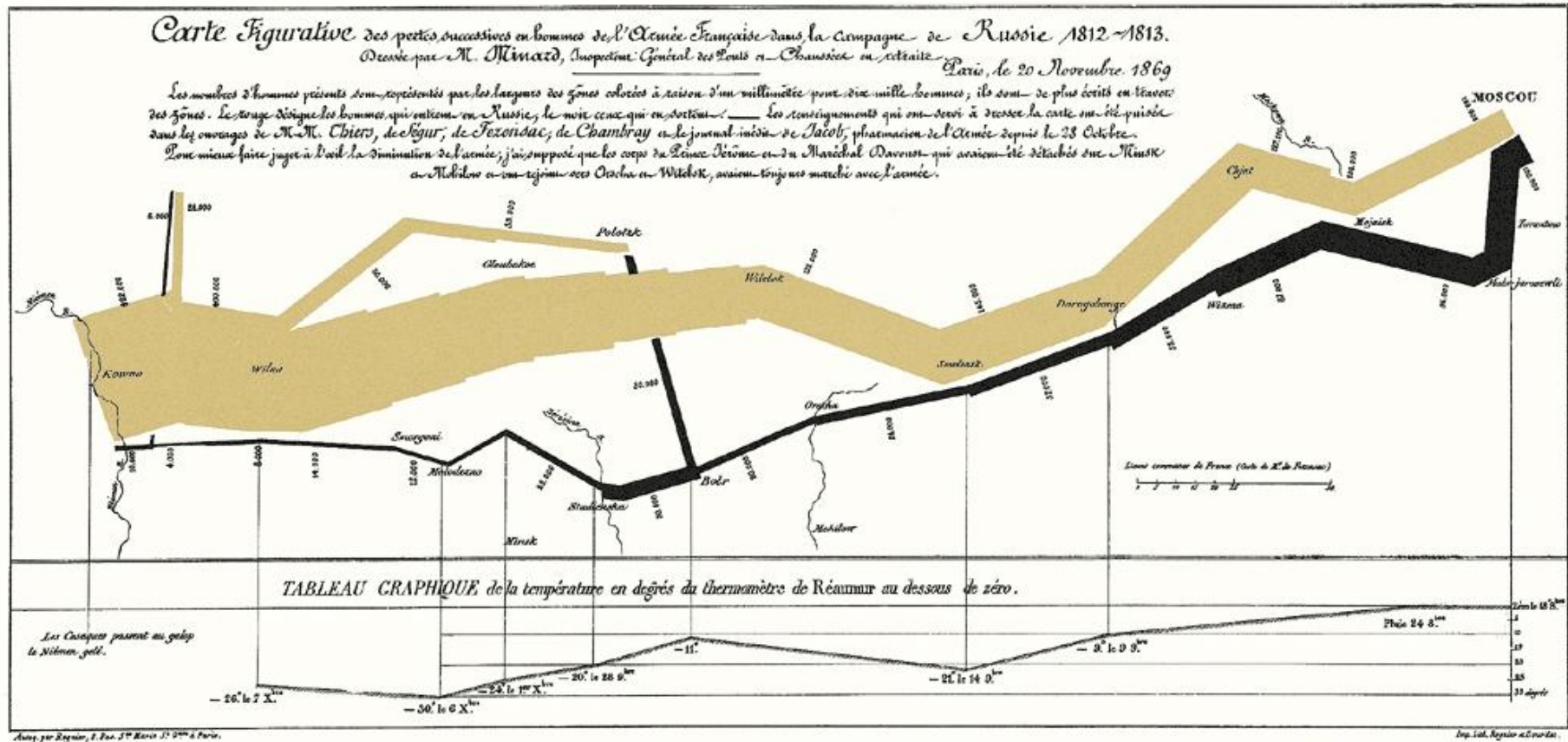
三维素描图



霍乱病例的分布

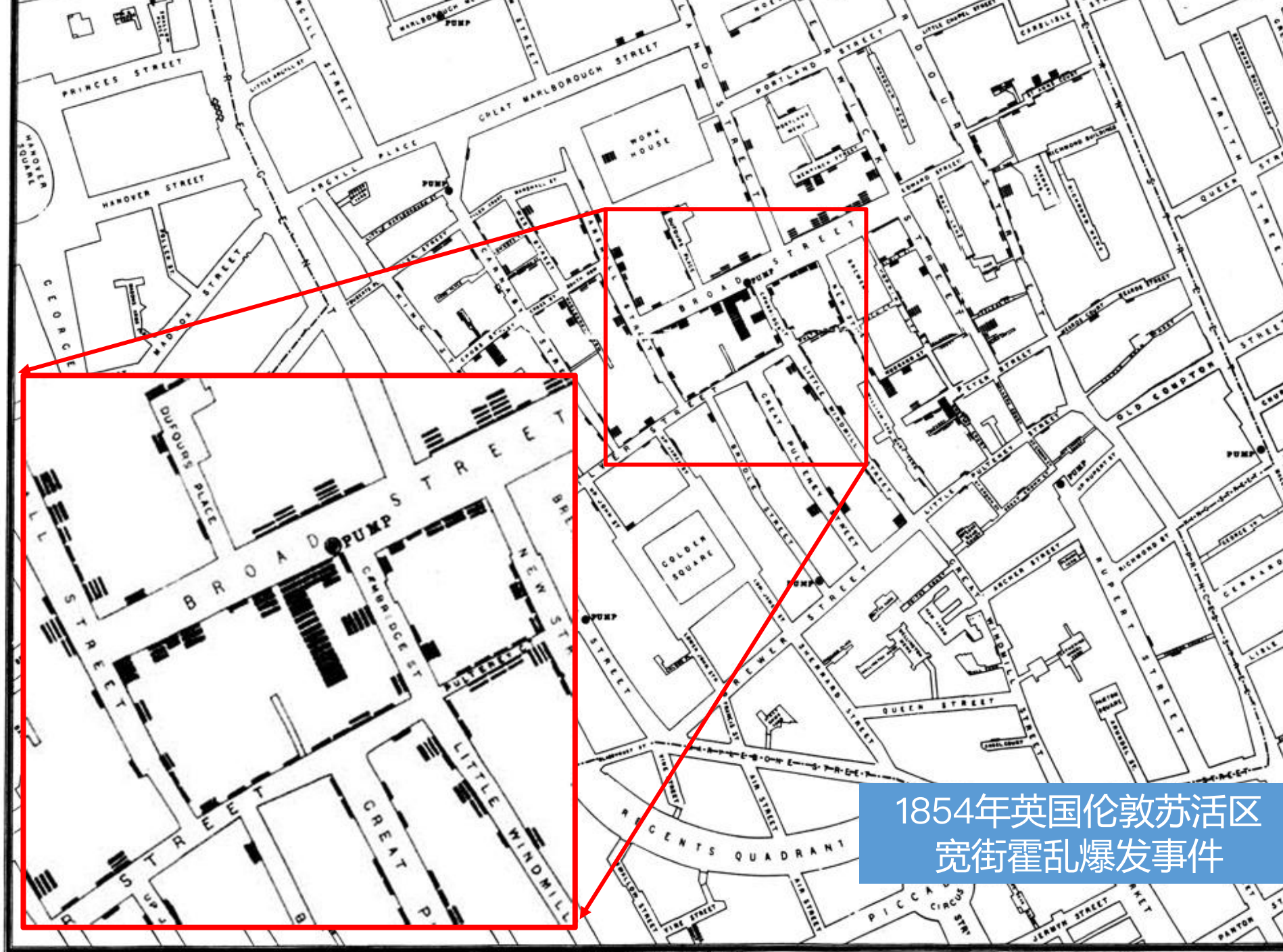


英国东征士兵死亡原因



C. J. Minard, 1869

E. Tufte, Writings, Artworks, News

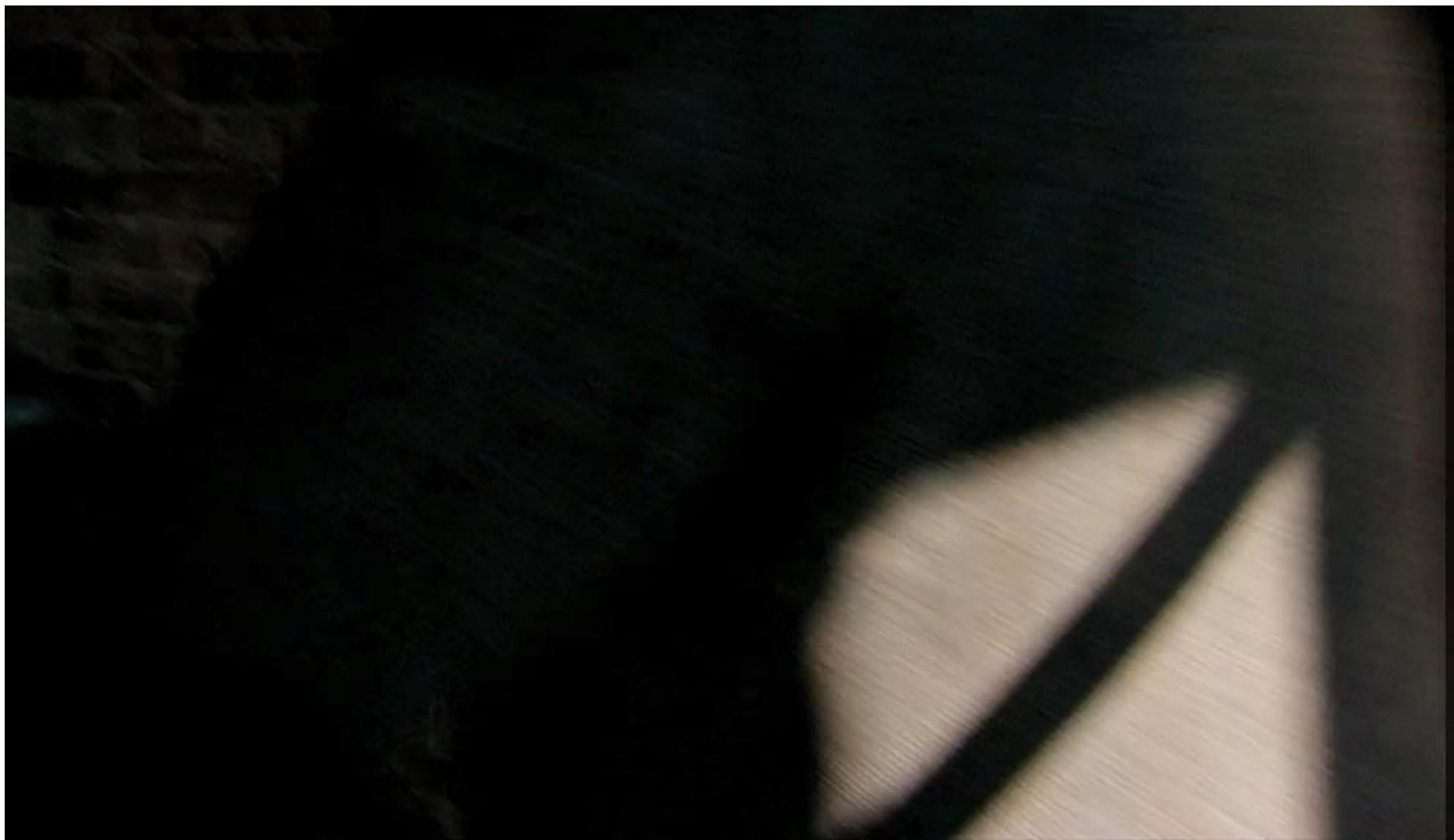


1854年英国伦敦苏活区
宽街霍乱爆发事件



浙江大學
ZHEJIANG UNIVERSITY

Hans Rosling' TED talk



为什么要用数据可视化？



为什么要用数据可视化

- 人类认知存在天生的缺陷
 - Change blindness
- 统计数据存在盲区
 - Anscombe's Quartet
- 可解释性的一种途径
 - LLM visualization

**Watch what happens as the
unsuspecting pedestrian
provides directions.**

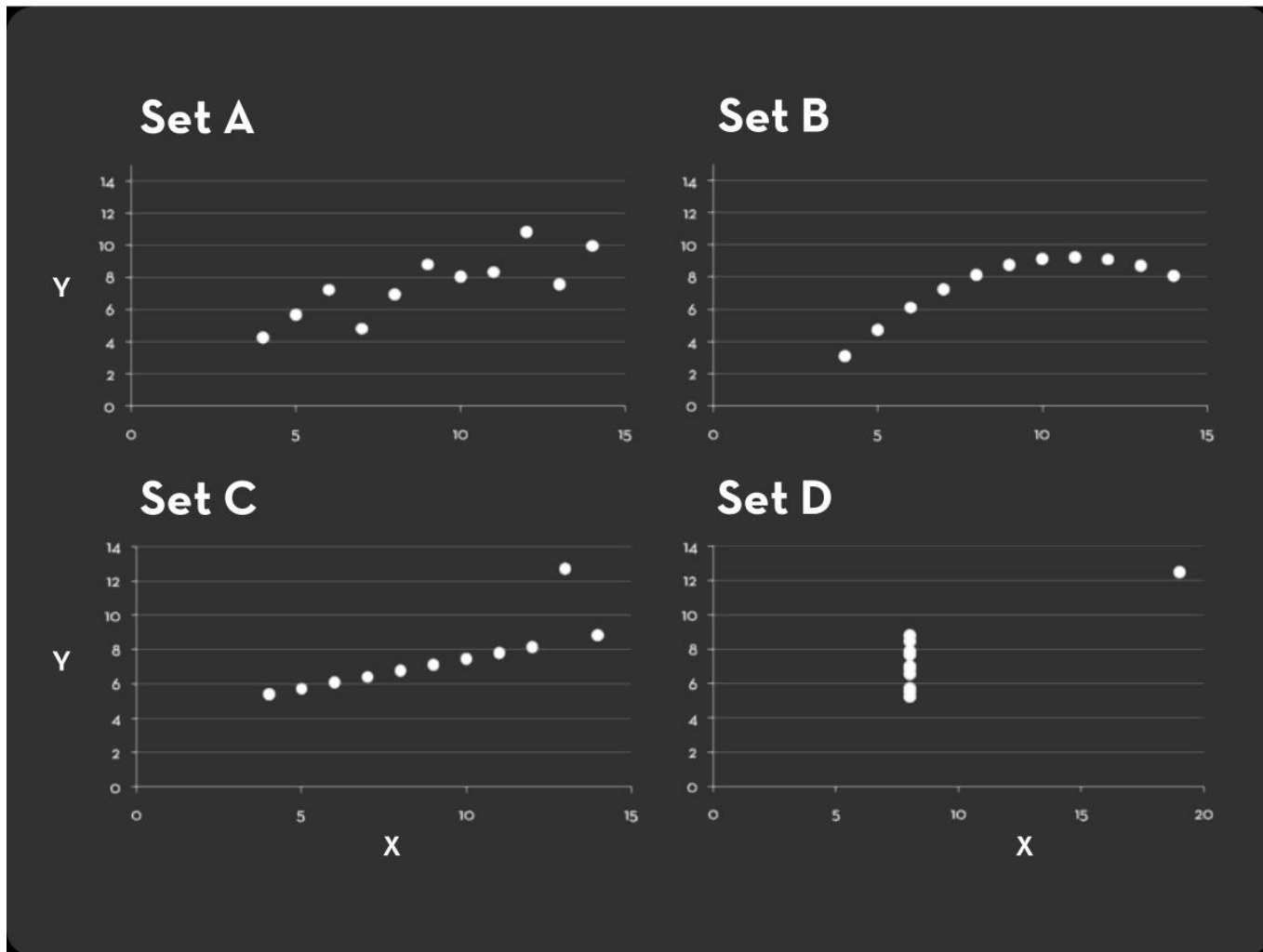
Daniel J. Simons and Daniel T. Levin,
Failure to detect changes to people during a real world interaction, 1998

统计数据的盲区



Set A		Set B		Set C		Set D	
X	Y	X	Y	X	Y	X	Y
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.11	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89
Summary Statistics				Linear Regression			
$\mu_X = 9.0 \quad \sigma_X = 3.317$				$Y = 3 + 0.5 X$			
$\mu_Y = 7.5 \quad \sigma_Y = 2.03$				$R^2 = 0.67$			
				[Anscombe 73]			

统计数据的盲区



小测试



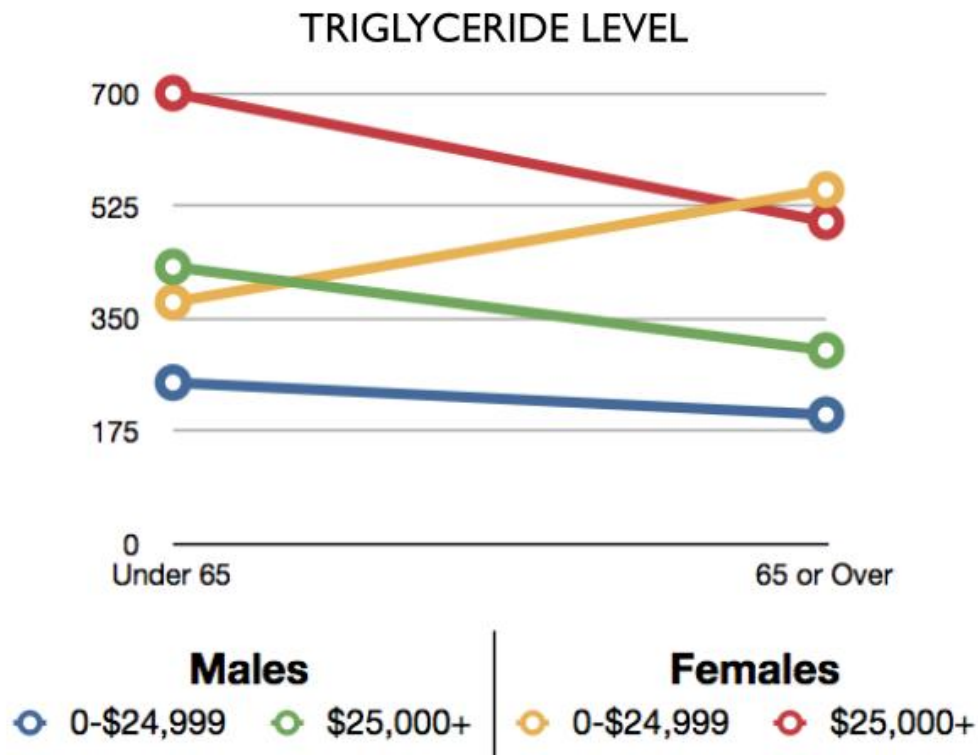
三酸甘油酯水平在**哪个年龄段和收入水平的人群**中
随年龄增长表现出不同的趋势？

	男性		女性	
收入水平	65岁以下	65岁及以上	65岁以下	65岁及以上
0-\$24,999	250	200	375	550
\$25,000+	430	300	700	500

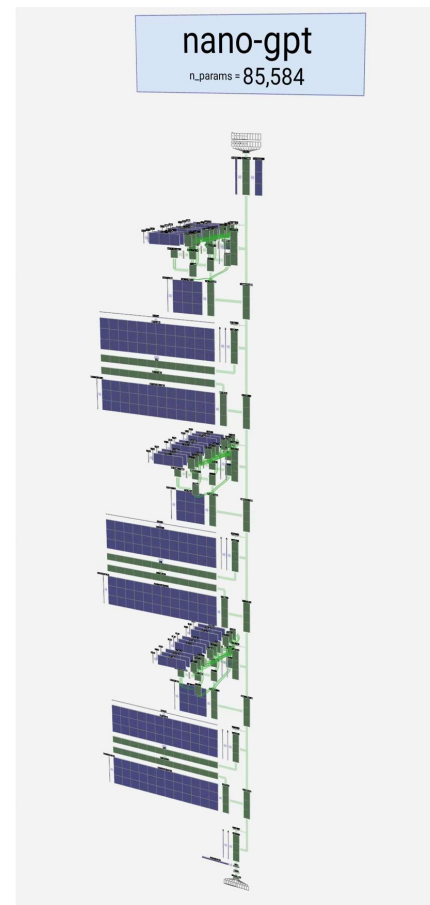
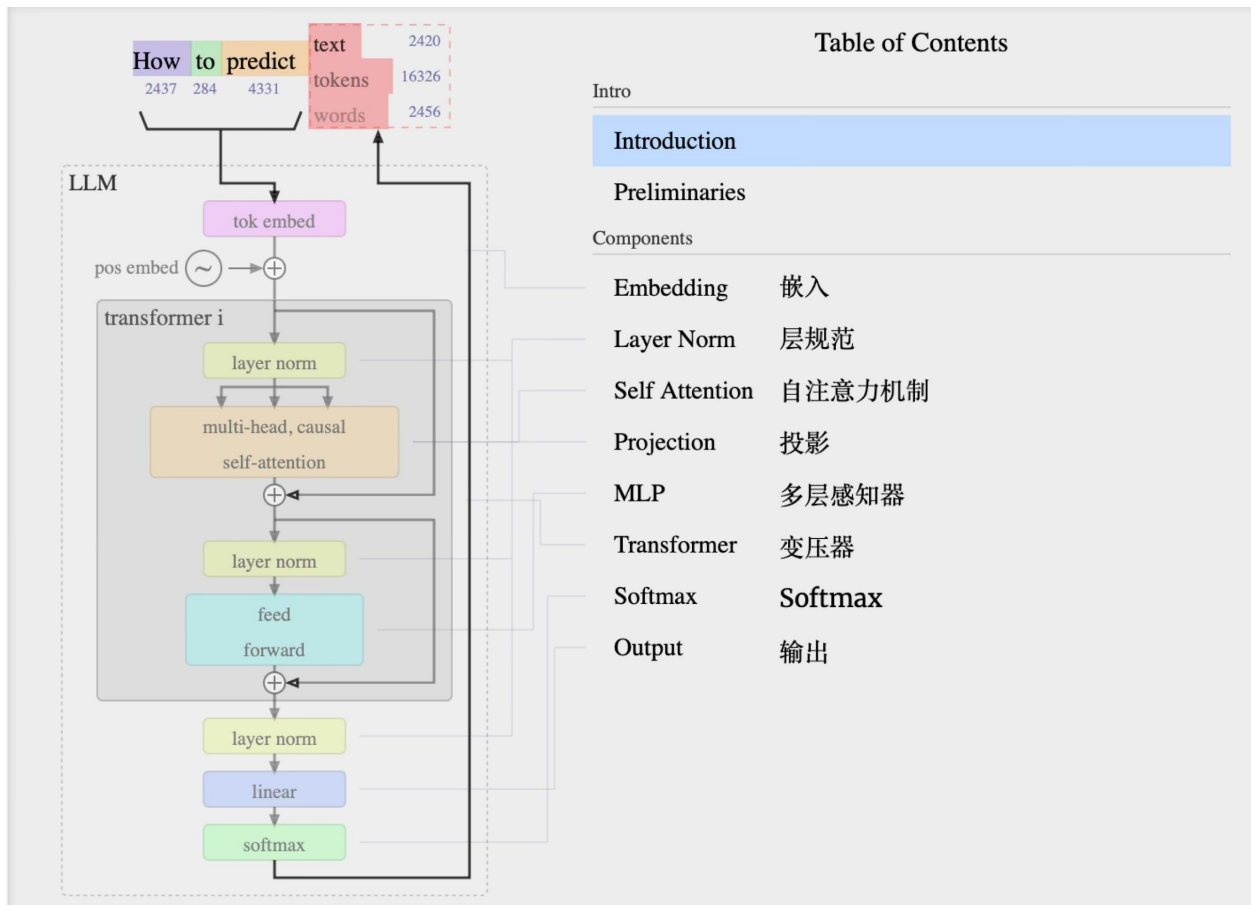
数据可视化的力量



三酸甘油酯水平在哪个年龄段和收入水平的人群中随年龄增长表现出不同的趋势？

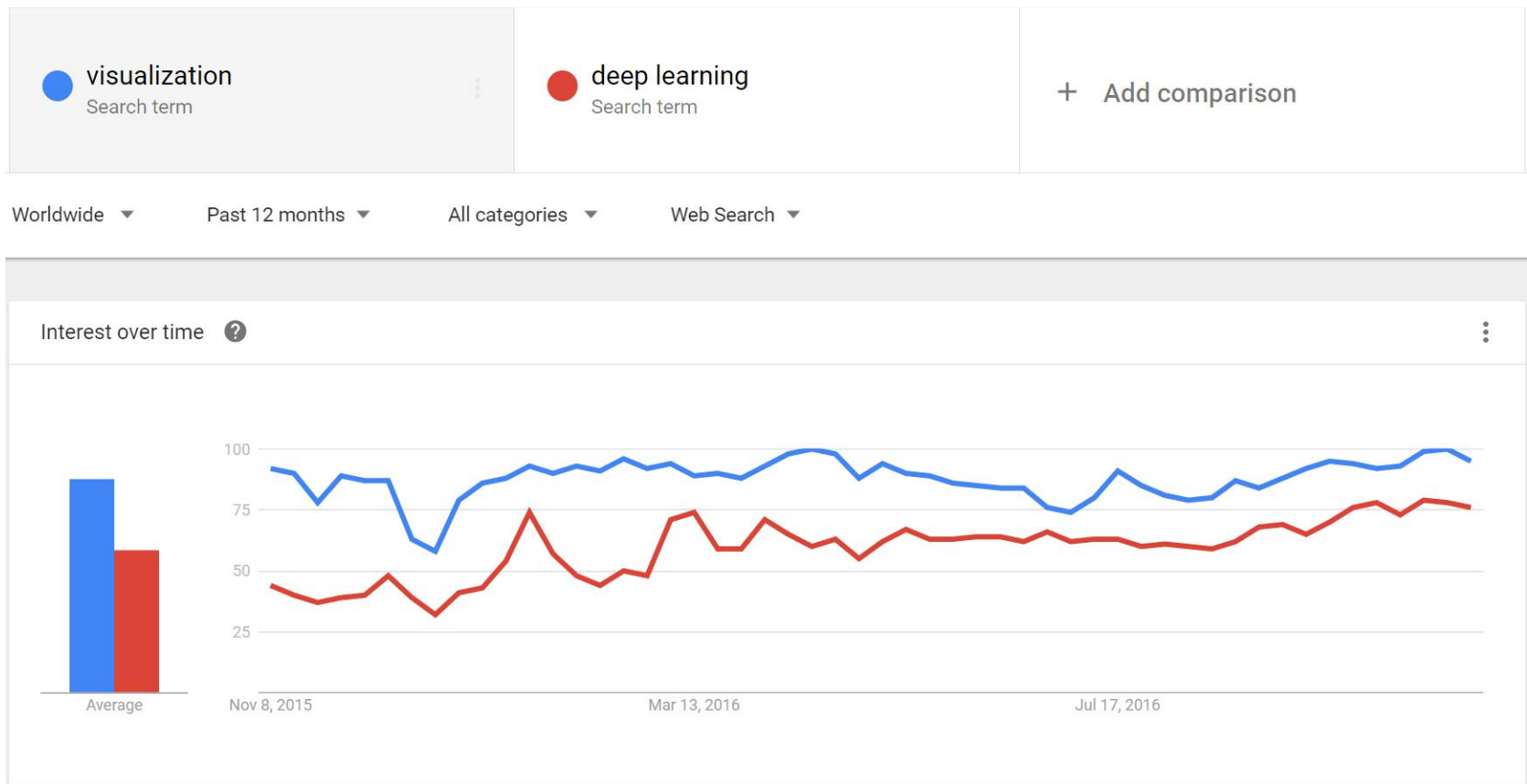


数据可视化与大模型长什么样？



<https://bbycroft.net/llm>

可视化与深度学习



可视化是一个跨学科领域



图形学

- 照片真实感
- 模拟
- 现实世界
- 视觉媒介

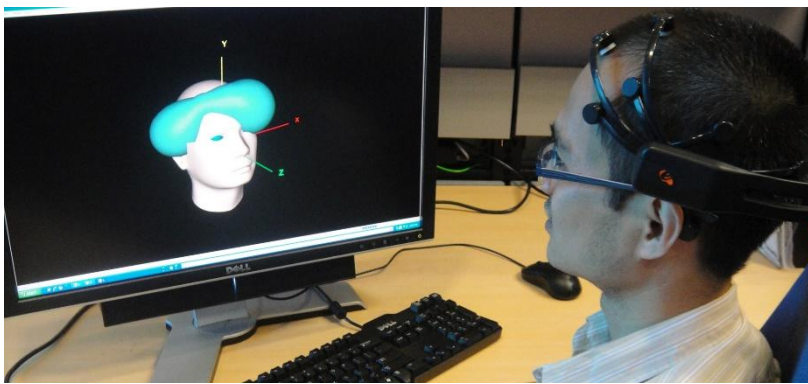
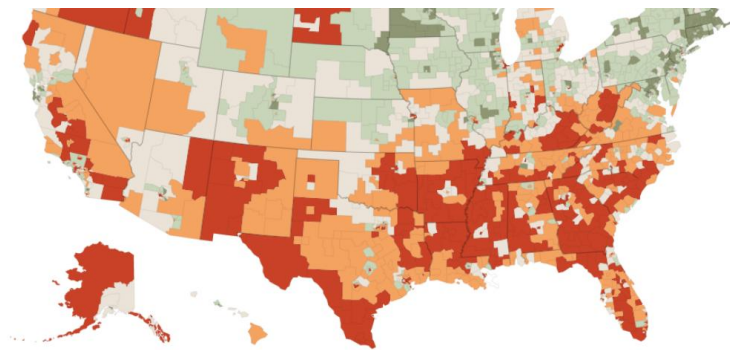
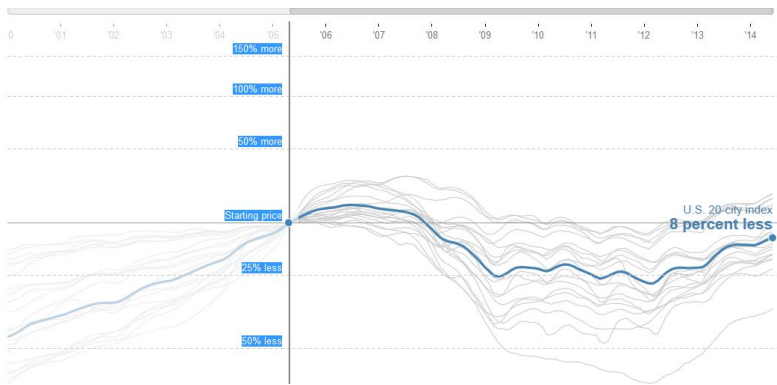


可视化

- 描述性
- 增强理解
- 传达信息

可视化与人机交互

- 可视化主要处理数据
- 人机交互主要关注人与机器之间的交互行为

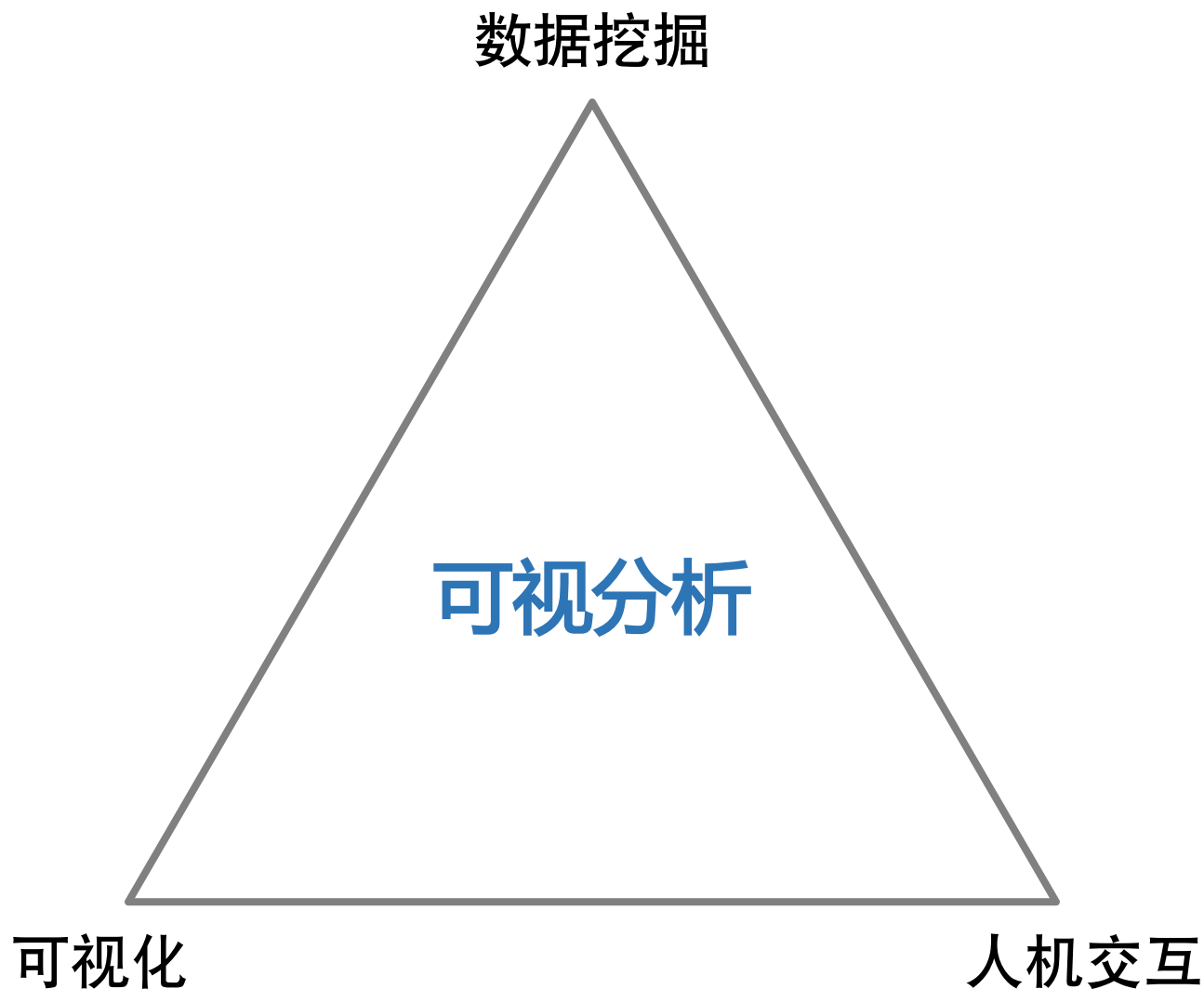


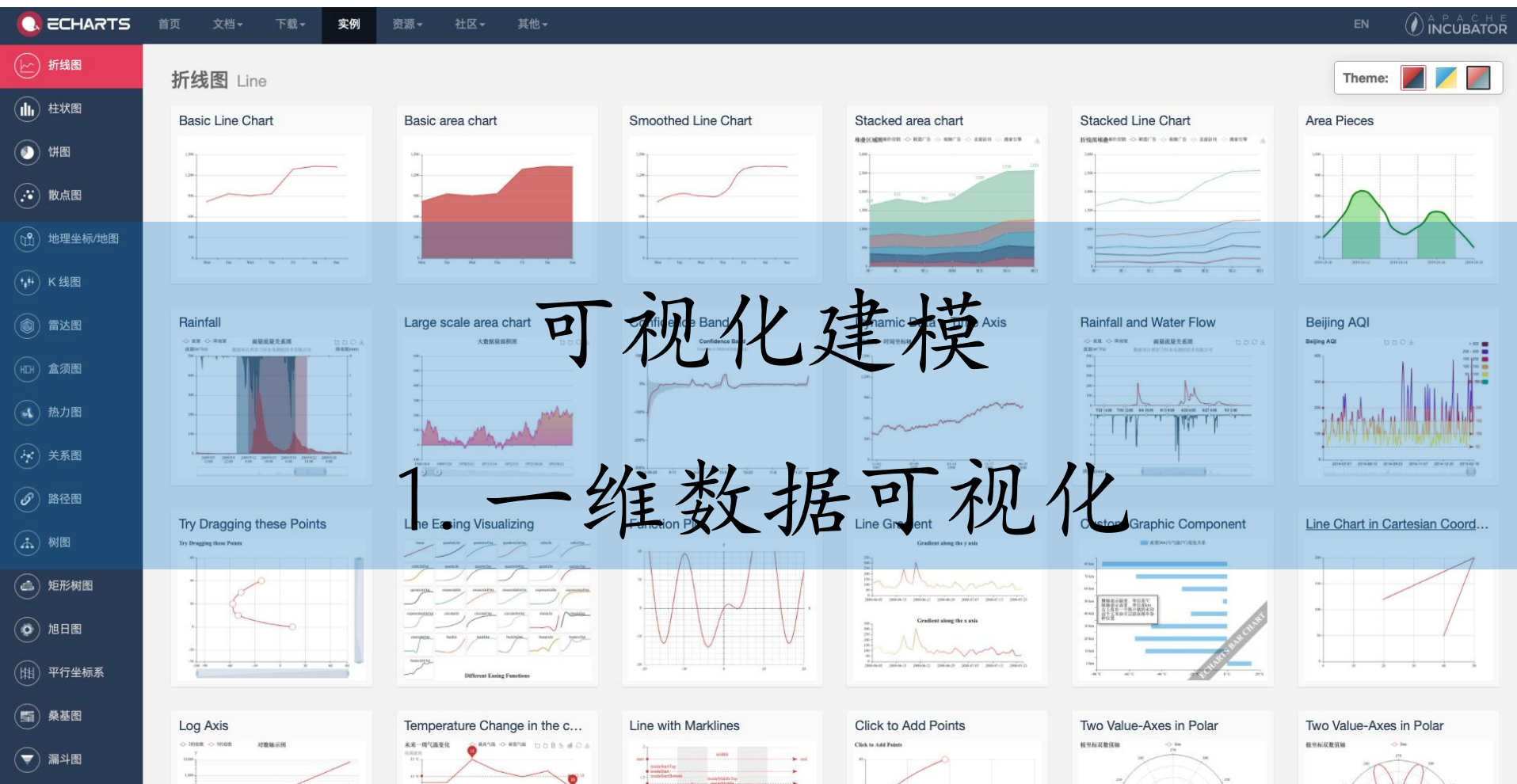
可视化与数据挖掘

- 数据挖掘 (Data Mining) 更关注自动化的算法
- 可视化通过让人类参与其中实现交互式的数据分析



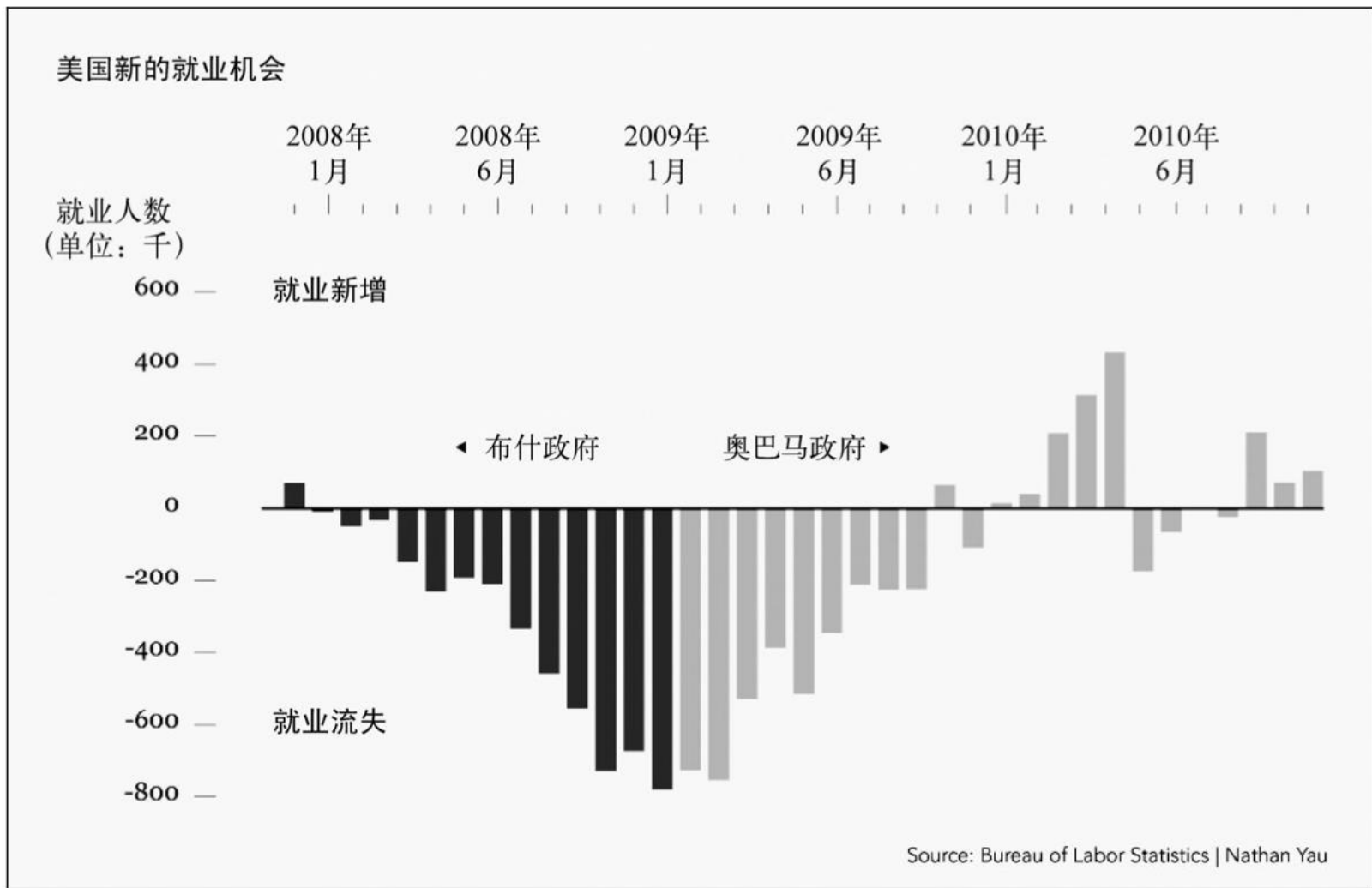
(港科大) 屈华民教授对可视化与数据挖掘的观点





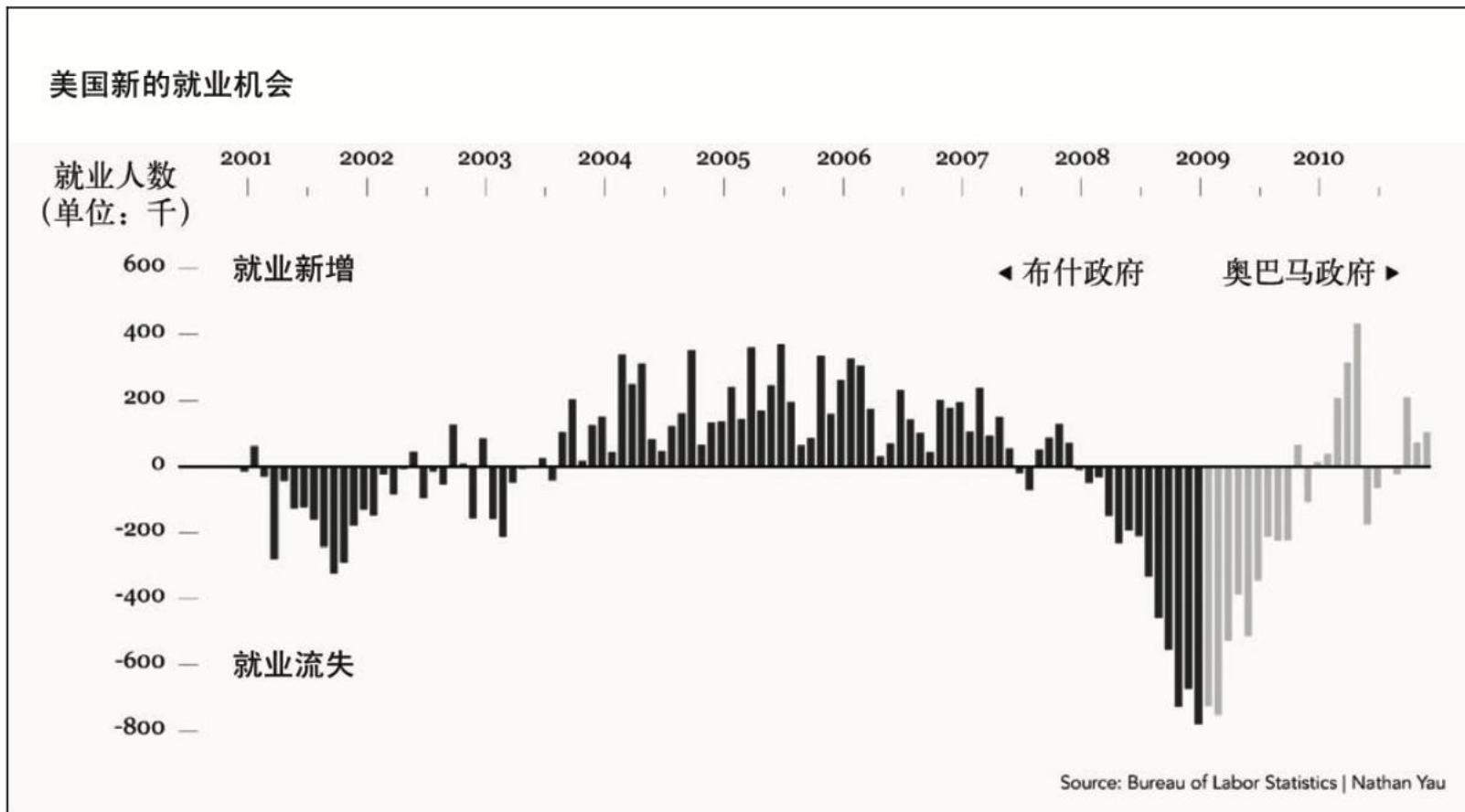
以Echarts为例 <https://echarts.apache.org/zh/index.html>

一维数据可视化



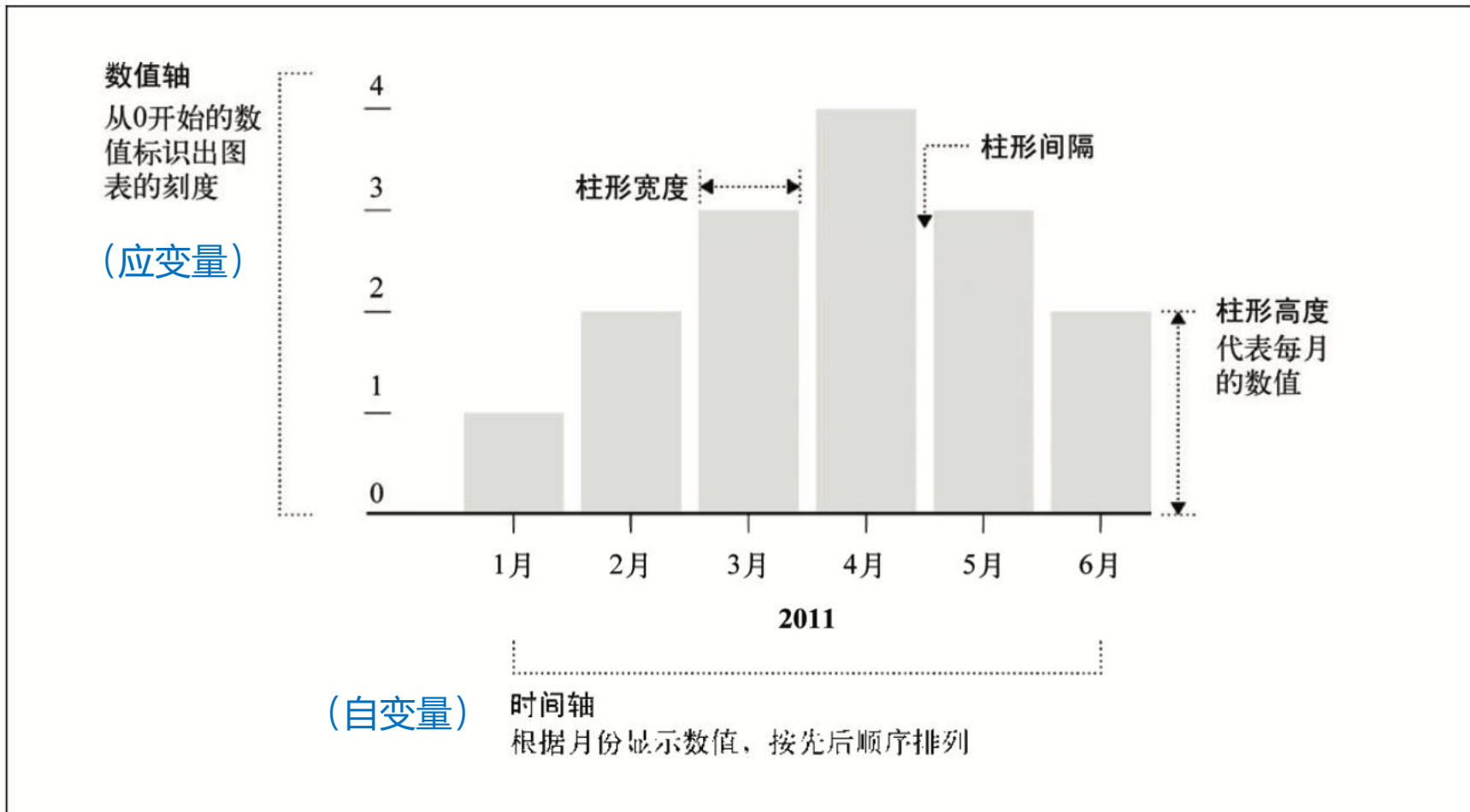
巴拉克·奥巴马执政后失业的变化

一维数据可视化



2001—2010年的美国失业变化

一维数据可视化 – 柱形图



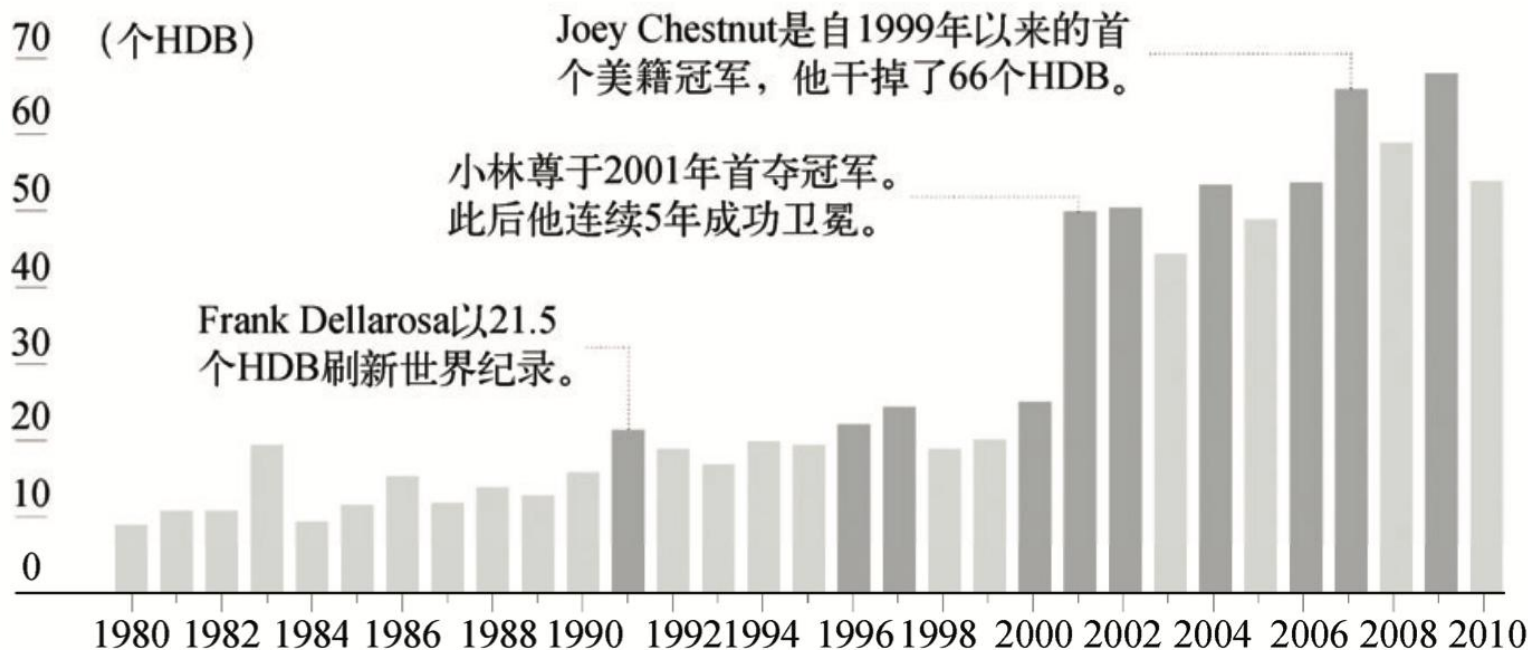
柱形图的基本框架



一维数据可视化 – 柱形图例

热狗大胃王

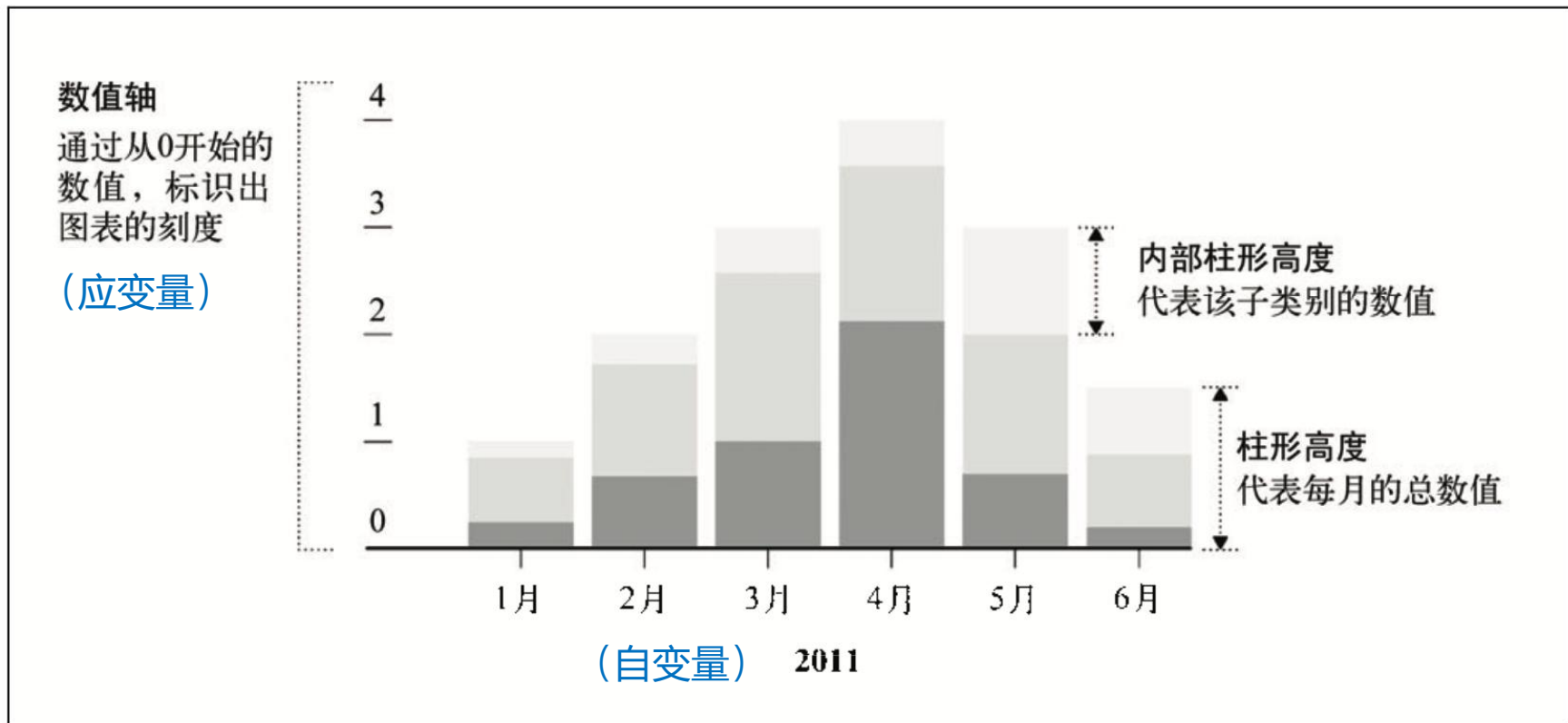
每年7月4日举办的“内森杯”热狗大胃王比赛始于20世纪初，但直到2001年才开始获得广泛关注。当时来自日本的选手小林尊横空出世，一举将该项世界纪录刷新至之前的两倍多。绿色显示的柱形表示新纪录。



Source: Wikipedia | Nathan Yau

“内森杯”热狗大胃王比赛成绩的柱形图

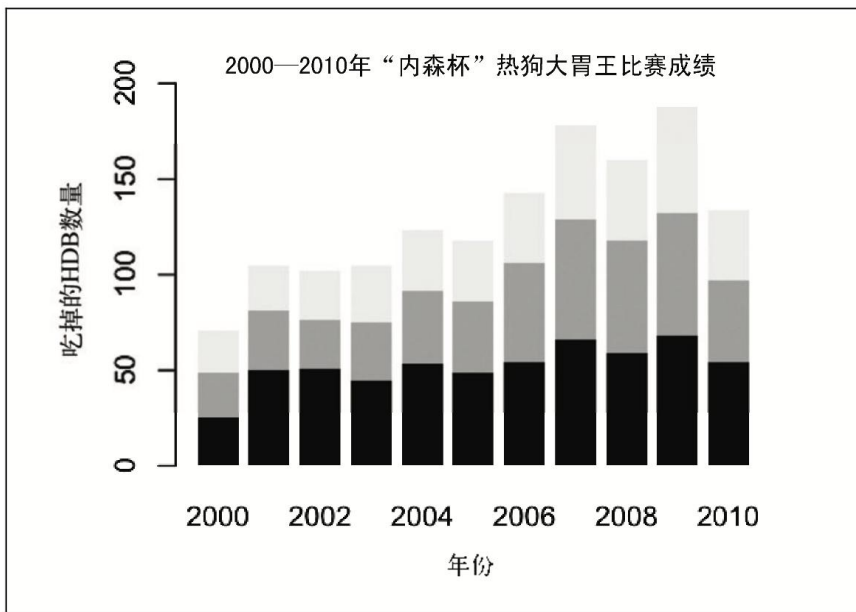
一维数据可视化 – 堆叠柱形图



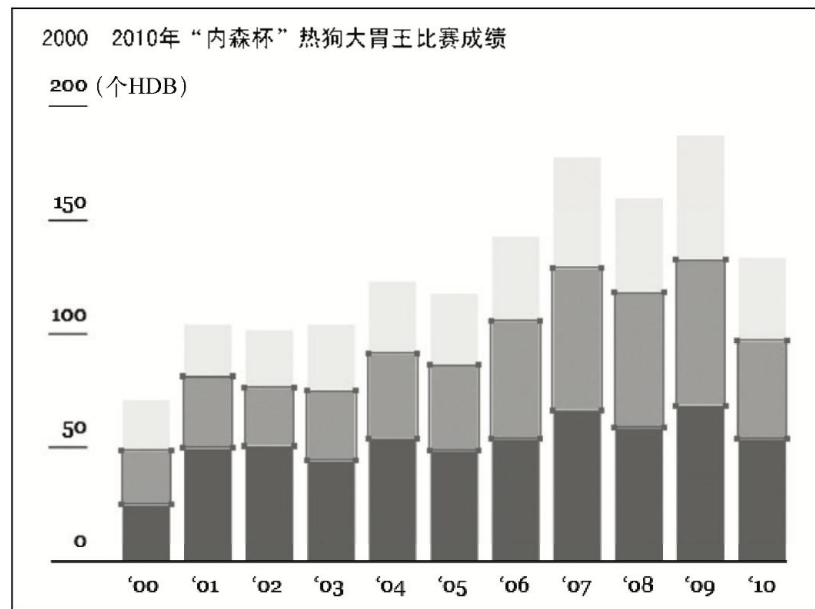
堆叠柱形图的基本框架



一维数据可视化 – 堆叠柱形图例

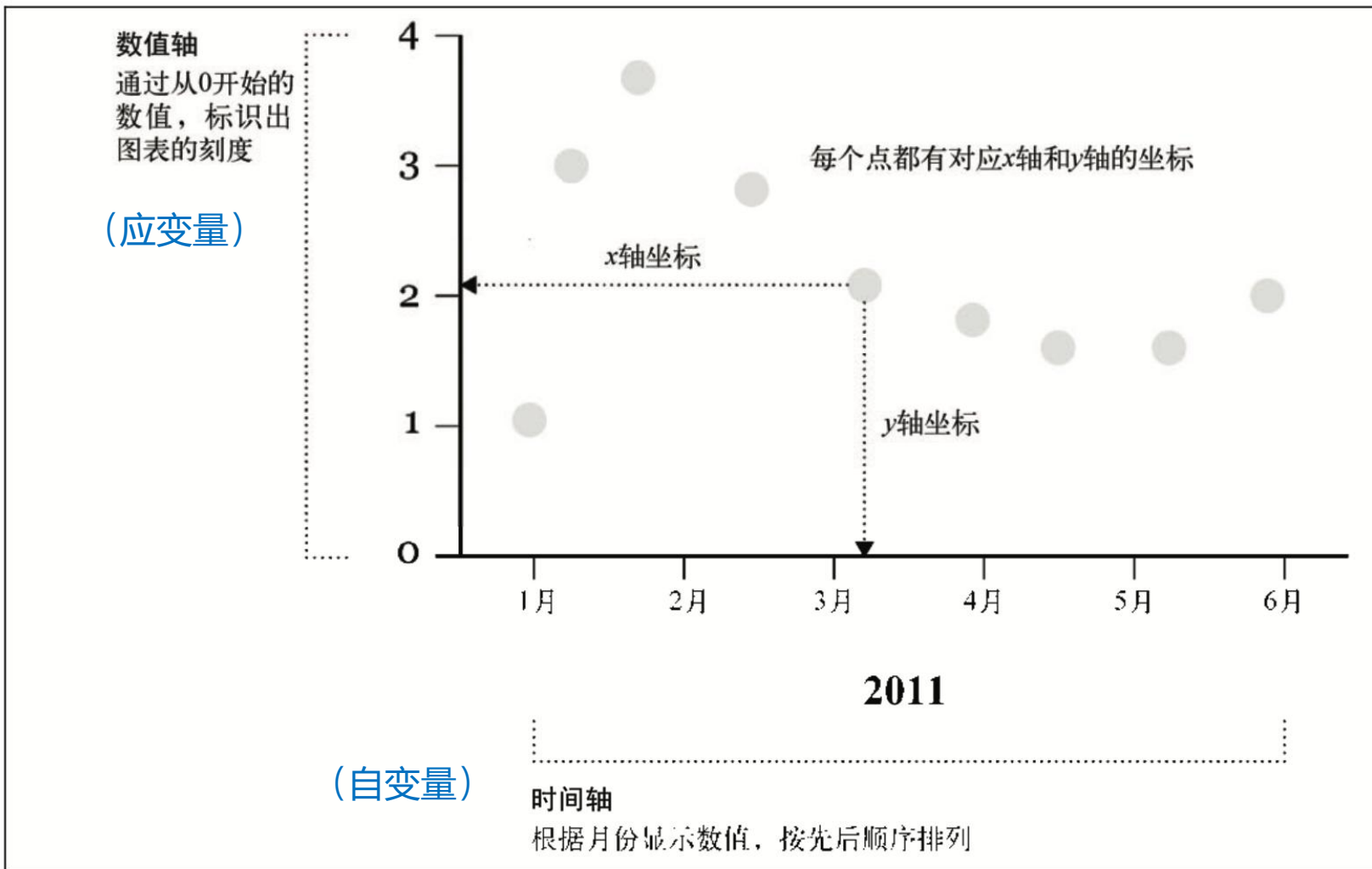


R语言创建的堆叠柱形图
(提示：调用`barplot`函数)



美化后的堆叠柱形图
(提示：使用`illustrator`)

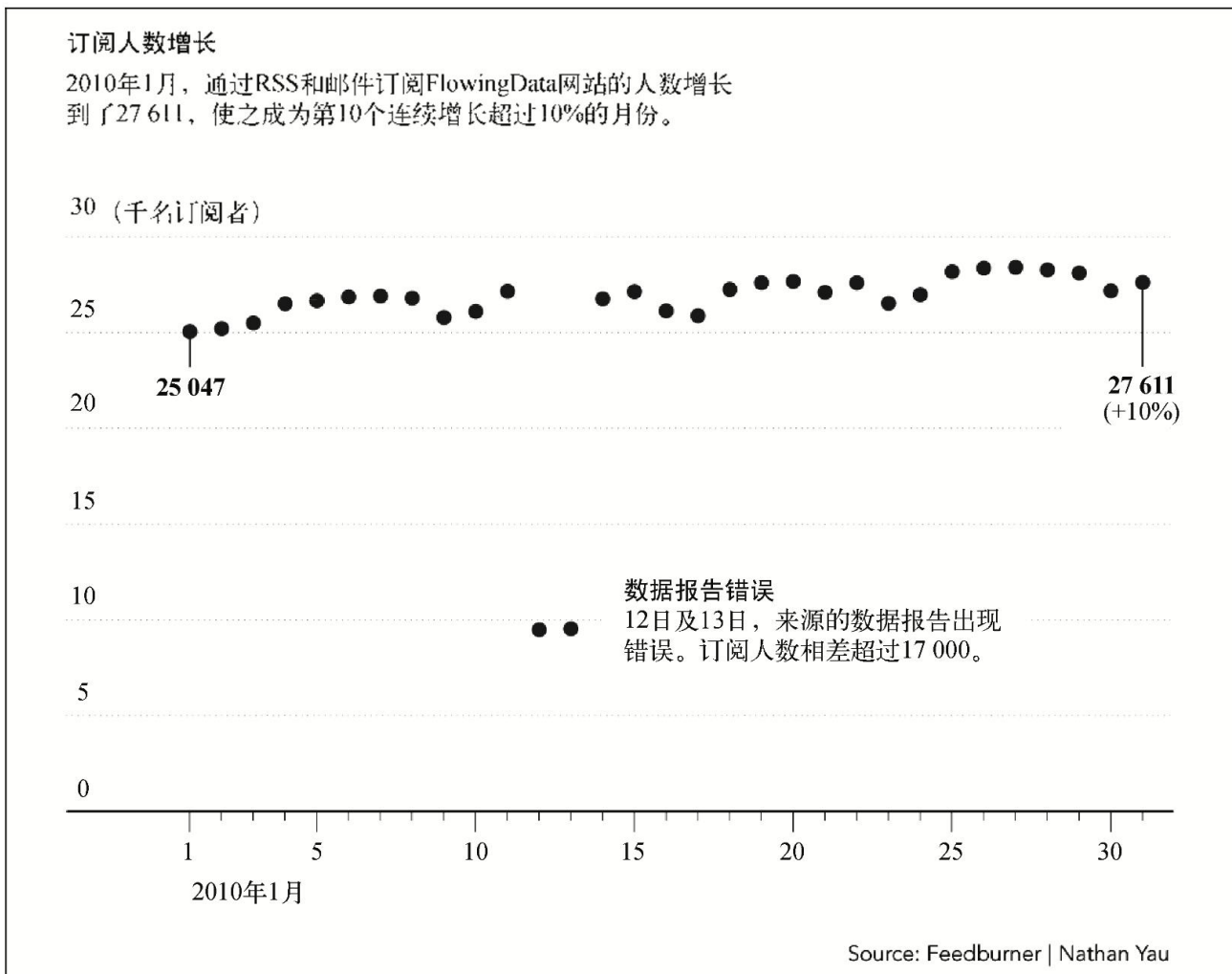
一维数据可视化 – 散点图



散点图的基本框架

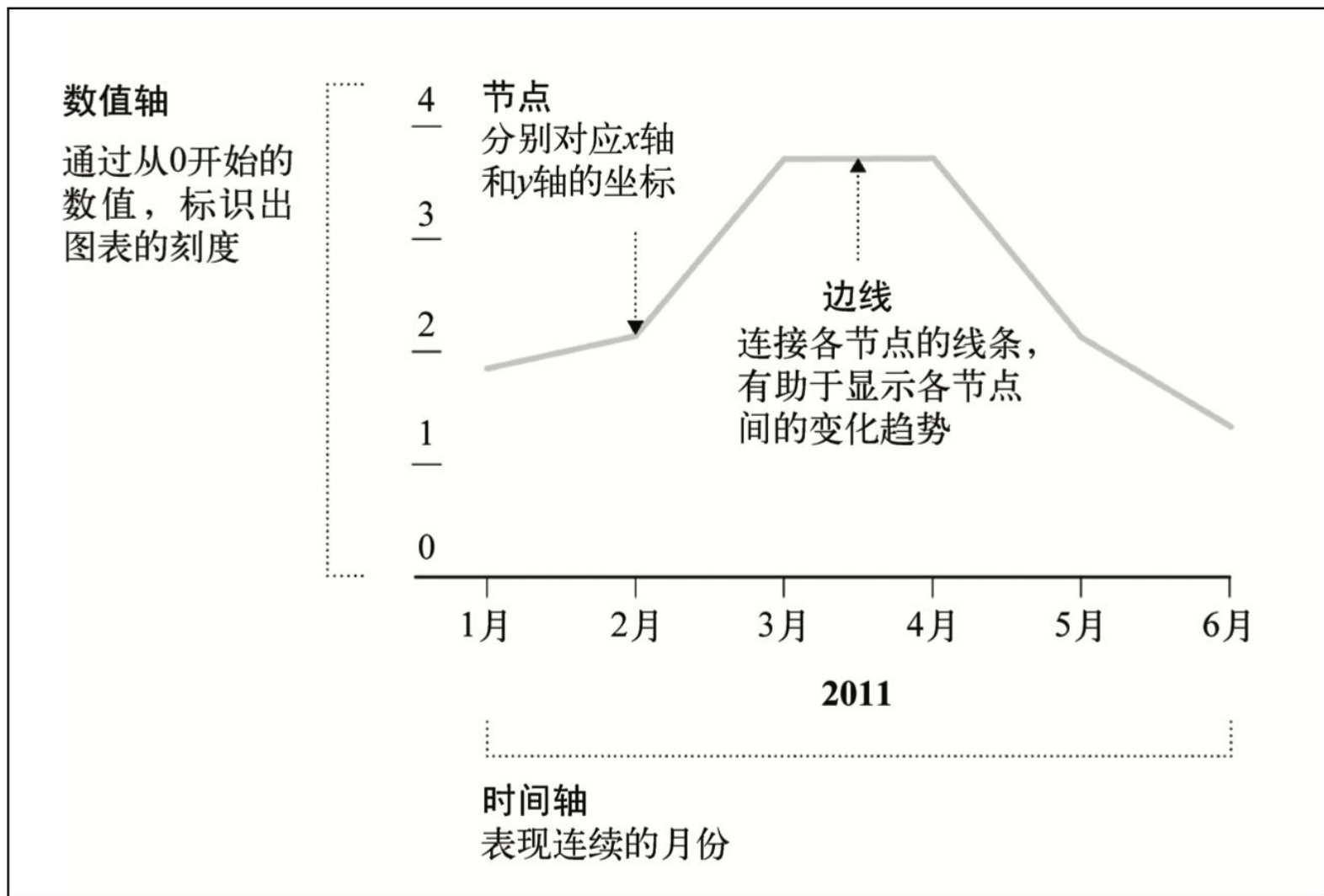


一维数据可视化 – 散点图例



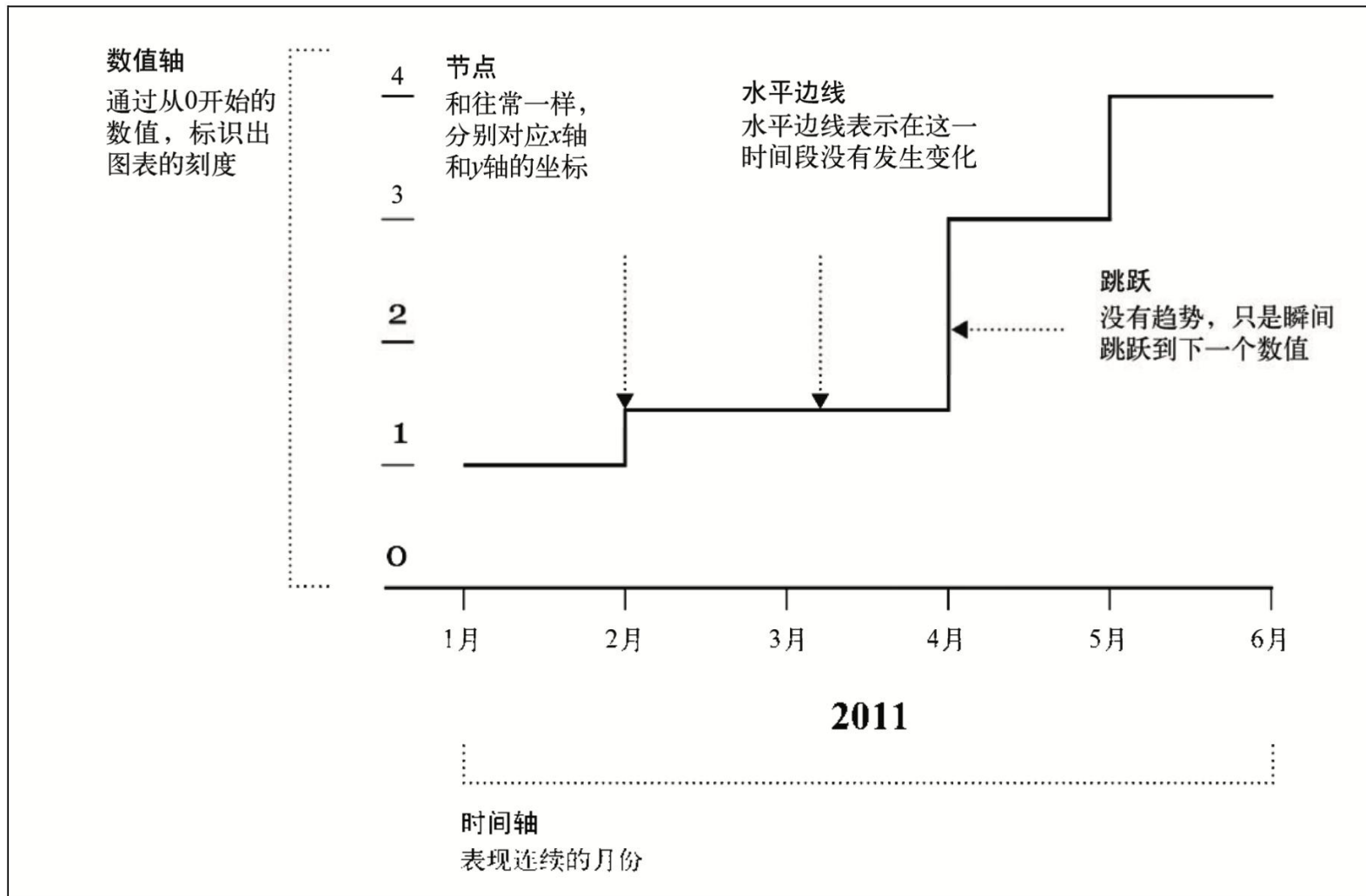
在R中创建并在Illustrator中设计得到的散点图

一维数据可视化 – 折线图



思考：延续型和离散型表达的异同点

一维数据可视化 – 阶梯图



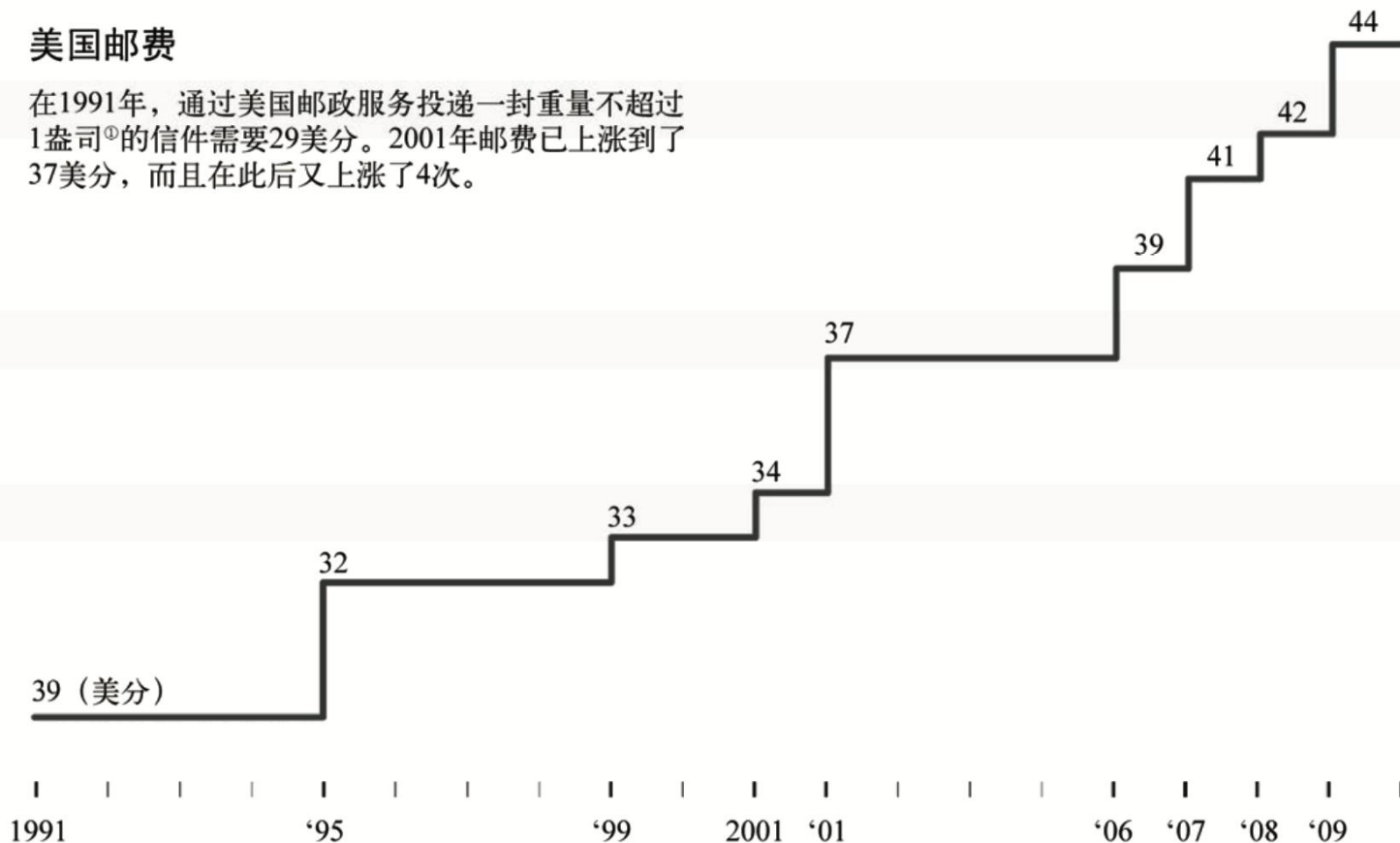
思考：延续型和离散型表达的异同点



一维数据可视化 – 阶梯图例

美国邮费

在1991年，通过美国邮政服务投递一封重量不超过1盎司[®]的信件需要29美分。2001年邮费已上涨到了37美分，而且在此后又上涨了4次。

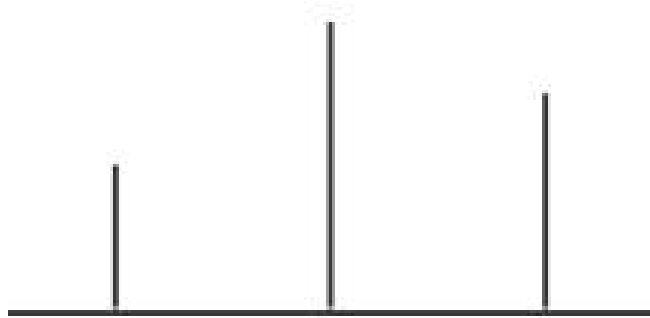


显示美国邮费变化的阶梯图

一维数据可视化 – 插值



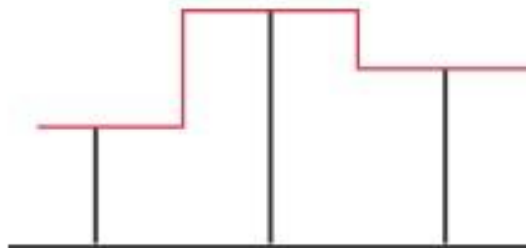
Goal: Interpolate Values



一维数据可视化 – 插值

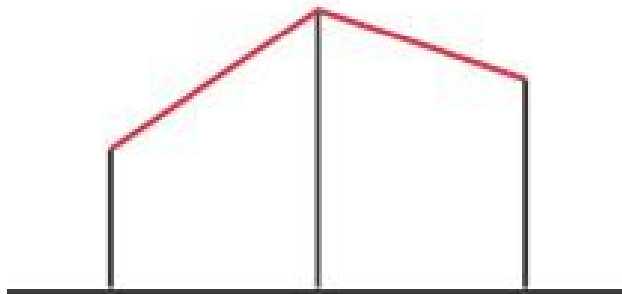


Nearest Neighbor Interpolation



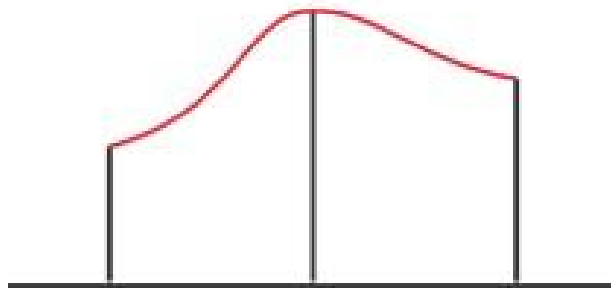
Problem: values not continuous

Linear Interpolation

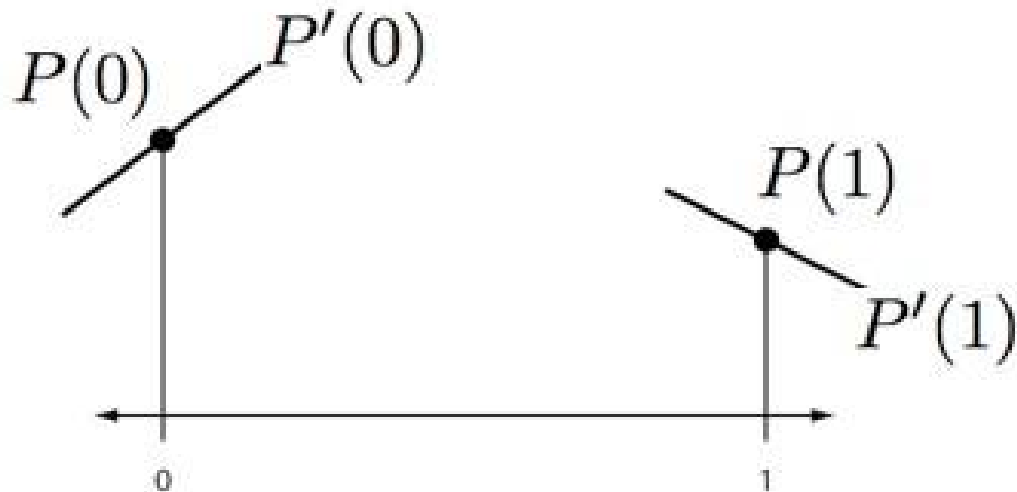


Problem: derivatives not continuous

Smooth Interpolation?



Cubic Hermite Interpolation



Given: values and derivatives at 2 points

Cubic Polynomial Interpolation

Assume cubic polynomial

$$P(t) = a t^3 + b t^2 + c t + d$$

Why? 4 constraints => need 4 degrees of freedom

Cubic Hermite Interpolation

Assume cubic polynomial

$$P(t) = at^3 + bt^2 + ct + d$$

$$P'(t) = 3at^2 + 2bt + c$$

Solve for coefficients:

$$P(0) = h_0 = d$$

$$P(1) = h_1 = a + b + c + d$$

$$P'(0) = h_2 = c$$

$$P'(1) = h_3 = 3a + 2b + c$$

Matrix Representation

$$h_0 = d$$

$$h_1 = a + b + c + d$$

$$h_2 = c$$

$$h_3 = 3a + 2b + c$$

$$\begin{bmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

一维数据可视化 – 插值

Matrix Representation of Polynomials

$$P(t) = \begin{bmatrix} a & b & c & d \end{bmatrix} \begin{bmatrix} t^3 \\ t^2 \\ t \\ 1 \end{bmatrix}$$



Hermite Basis Functions

$$\begin{bmatrix} a & b & c & d \end{bmatrix} \begin{bmatrix} t^3 \\ t^2 \\ t \\ 1 \end{bmatrix} = \begin{bmatrix} h_0 & h_1 & h_2 & h_3 \end{bmatrix} \begin{bmatrix} H_0(t) \\ H_1(t) \\ H_2(t) \\ H_3(t) \end{bmatrix}$$

$$P(t) = \sum_{i=0}^3 h_i H_i(t)$$

Matrix Representation

$$\begin{bmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$



Solve for a, b, c, d

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} h_0 \\ h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

Inverse Matrix

一维数据可视化 – 插值



Matrix Inverse

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Change Basis

$$\begin{bmatrix} a & b & c & d \end{bmatrix} \underbrace{\begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}}_{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}} \begin{bmatrix} t^3 \\ t^2 \\ t \\ 1 \end{bmatrix}$$

Change Basis

$$\begin{bmatrix} a & b & c & d \end{bmatrix} \underbrace{\begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}}_{\begin{bmatrix} h_0 & h_1 & h_2 & h_3 \end{bmatrix}} \begin{bmatrix} t^3 \\ t^2 \\ t \\ 1 \end{bmatrix}$$

一维数据可视化 – 插值



Change Basis

$$\underbrace{\begin{bmatrix} a & b & c & d \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}}_{\begin{bmatrix} h_0 & h_1 & h_2 & h_3 \end{bmatrix}} \underbrace{\begin{bmatrix} 2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^3 \\ t^2 \\ t \\ 1 \end{bmatrix}}_{\begin{bmatrix} H_0(t) \\ H_1(t) \\ H_2(t) \\ H_3(t) \end{bmatrix}}$$

一维数据可视化 – 插值

Hermite Basis Functions

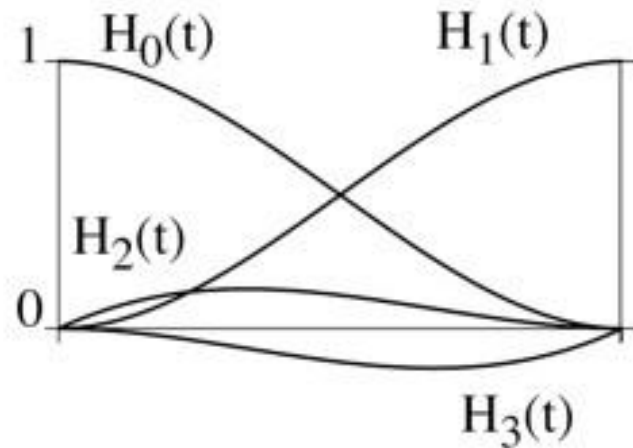
$$\begin{bmatrix} H_0(t) \\ H_1(t) \\ H_2(t) \\ H_3(t) \end{bmatrix} = \begin{bmatrix} 2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^3 \\ t^2 \\ t \\ 1 \end{bmatrix}$$

$$H_0(t) = 2t^3 - 3t^2 + 1$$

$$H_1(t) = -2t^3 + 3t^2$$

$$H_2(t) = t^3 - 2t^2 + t$$

$$H_3(t) = t^3 - t^2$$



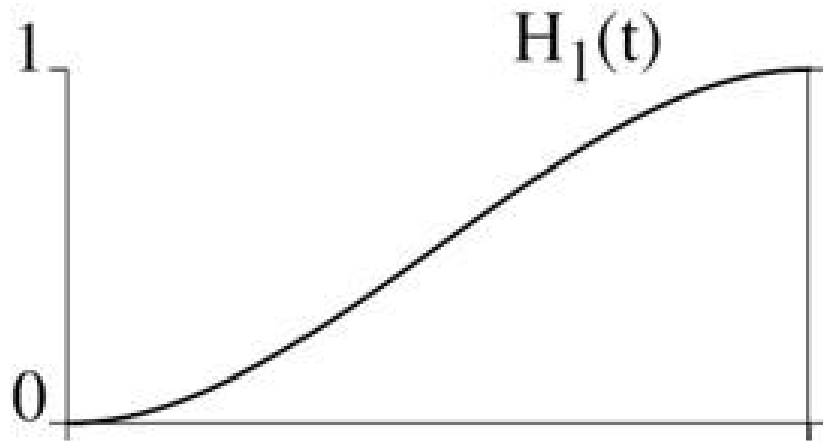
一维数据可视化 – 插值



Ease

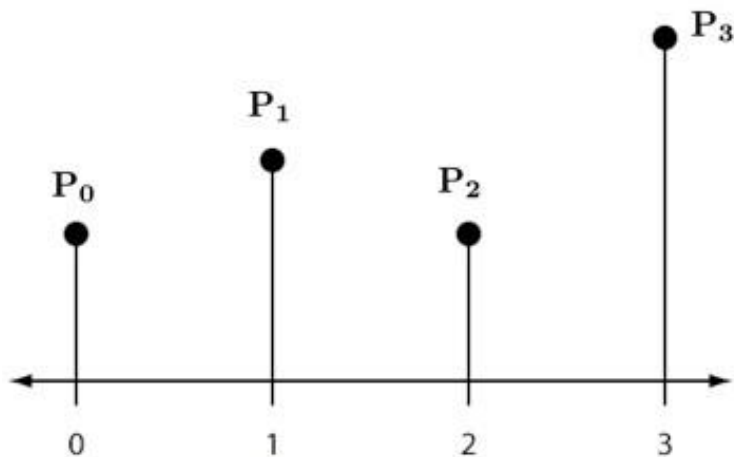
A very useful function

In animation, start and stop slowly (zero velocity)

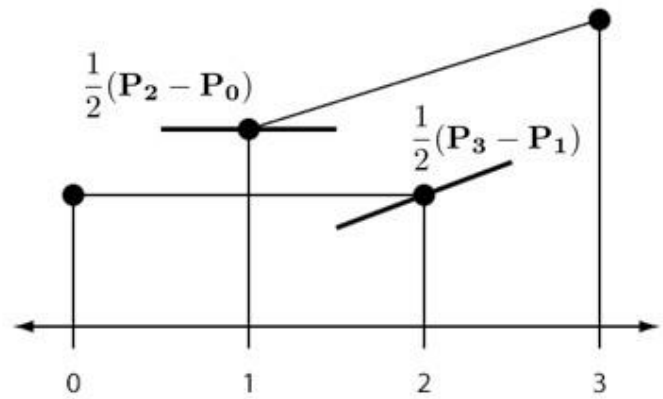
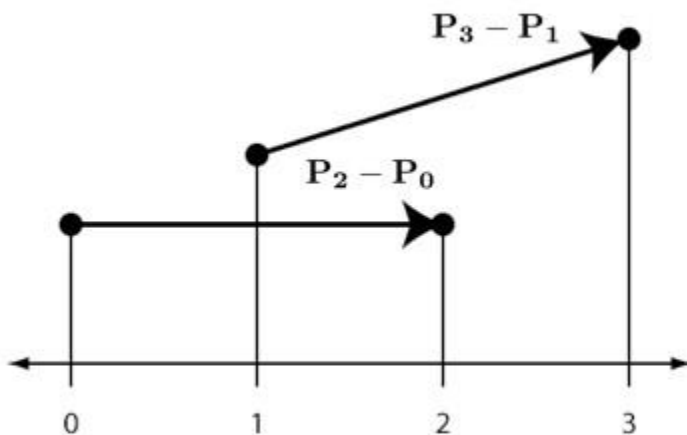
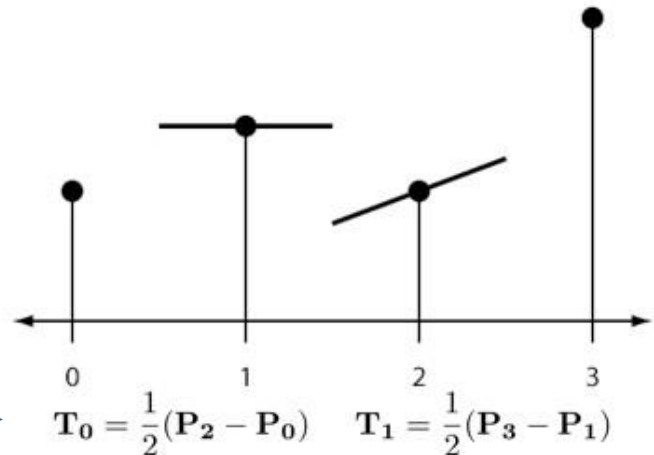


$$H_1(t) = -2t^3 + 3t^2 = t^2(3 - 2t)$$

一维数据可视化 – Catmull-Rom插值



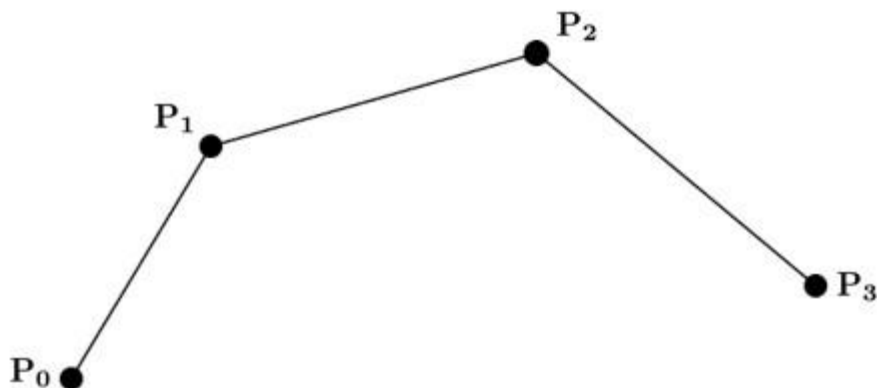
Interpolate points smoothly
Slopes not given though



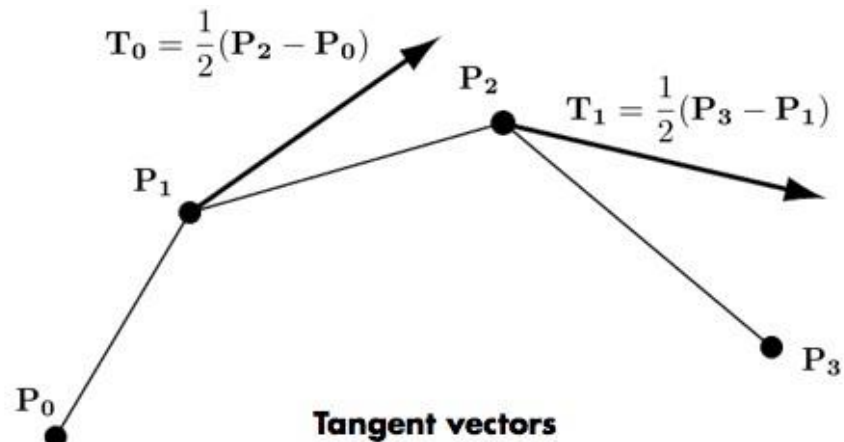
一维数据可视化 – Catmull-Rom插值



Catmull-Rom Interpolation

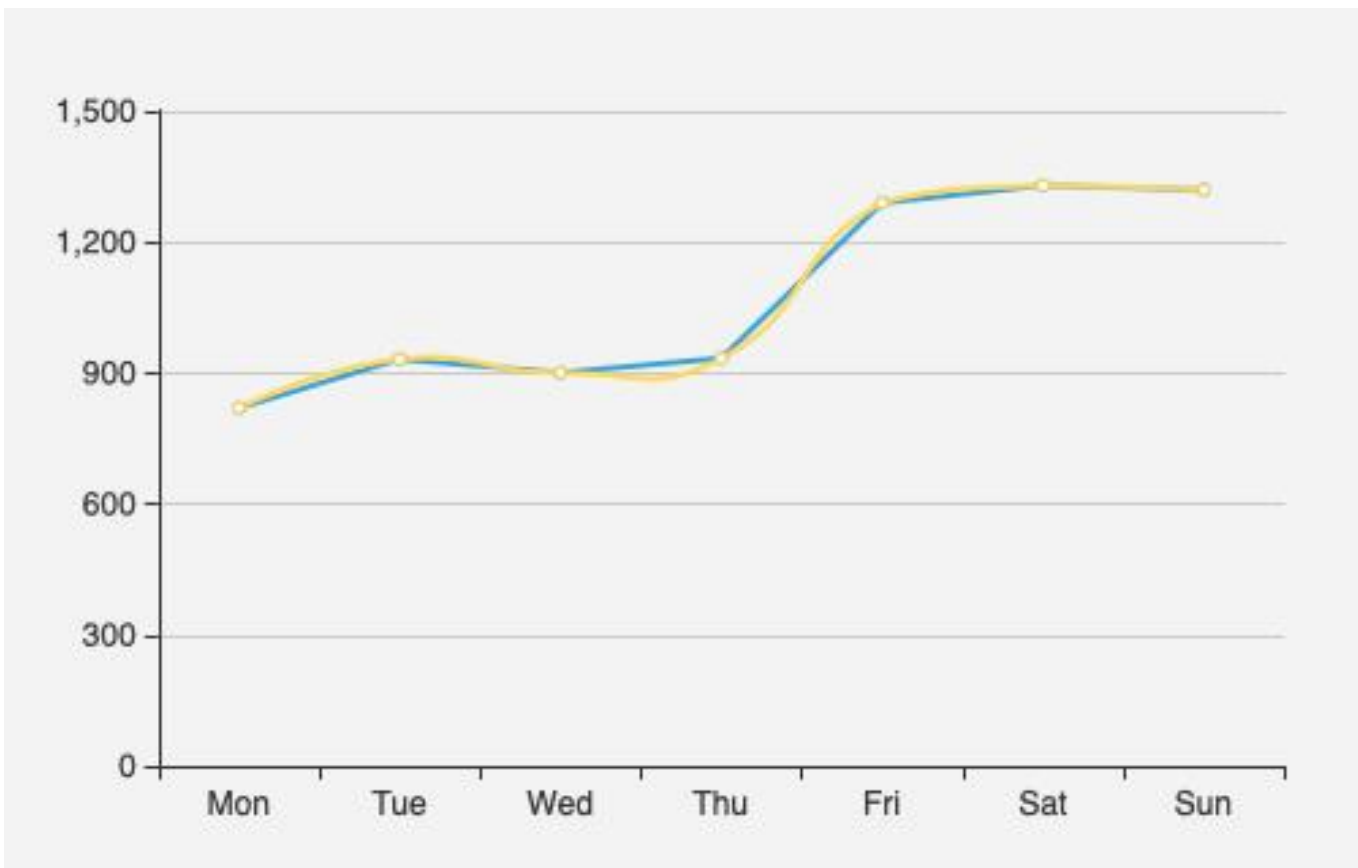


We can interpolate points as easily as values



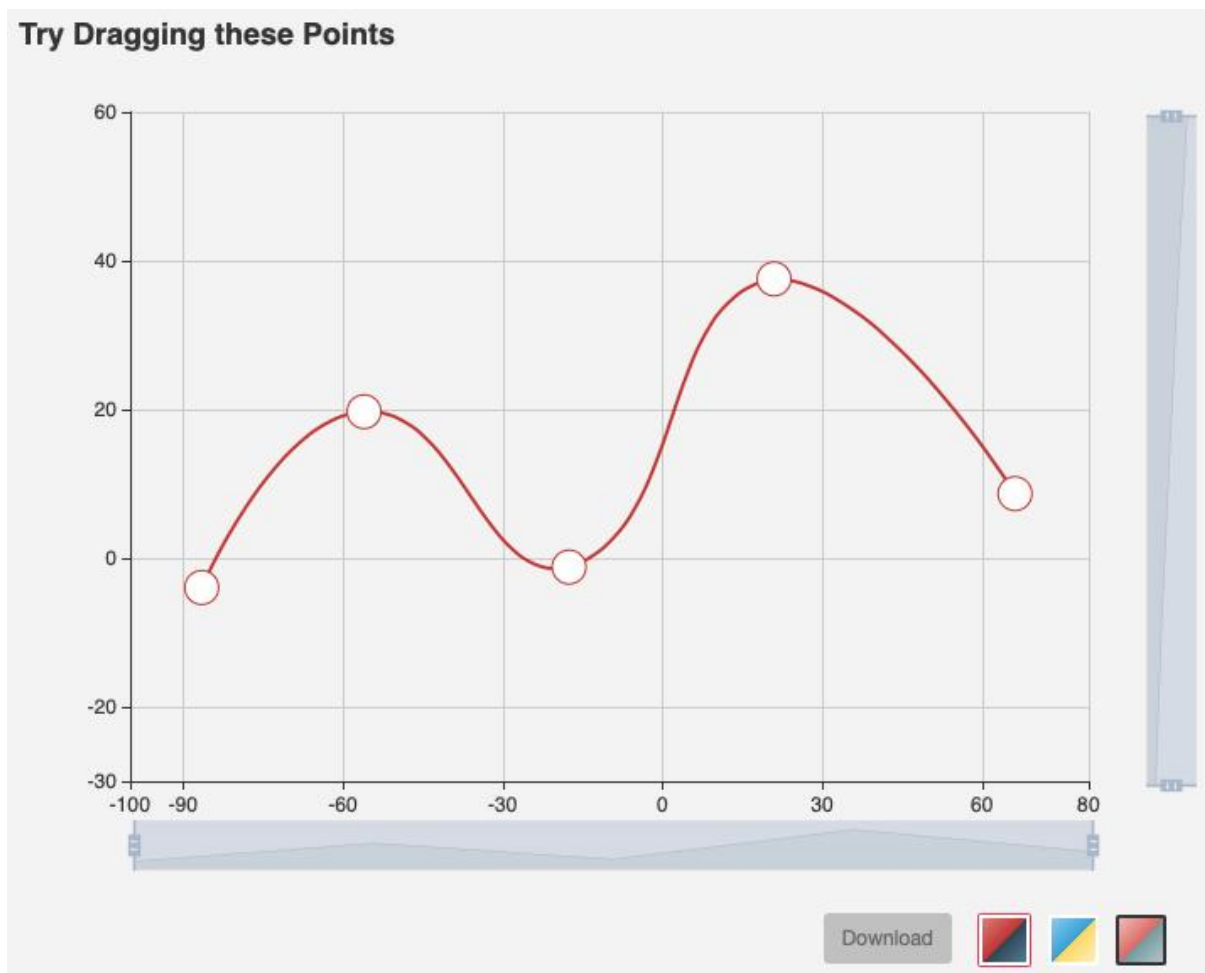
$$p(t) = (2t^3 - 3t^2 + 1)p_0 + (t^3 - 2t^2 + t)m_0 + (-2t^3 + 3t^2)p_1 + (t^3 - t^2)m_1$$

一维数据可视化 – Catmull-Rom插值例



Echarts中的例子

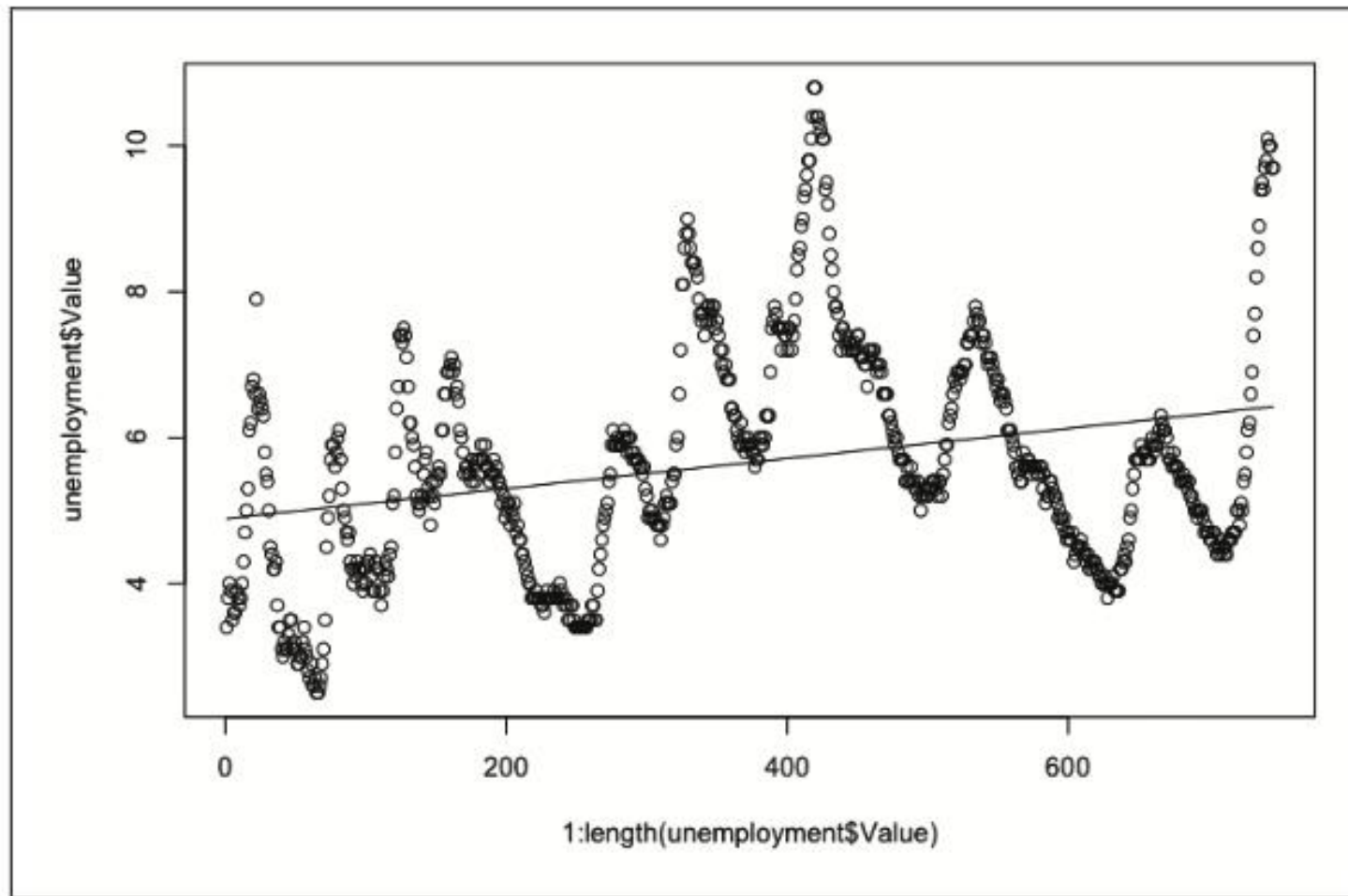
一维数据可视化 – Catmull-Rom插值例



Echarts中的例子

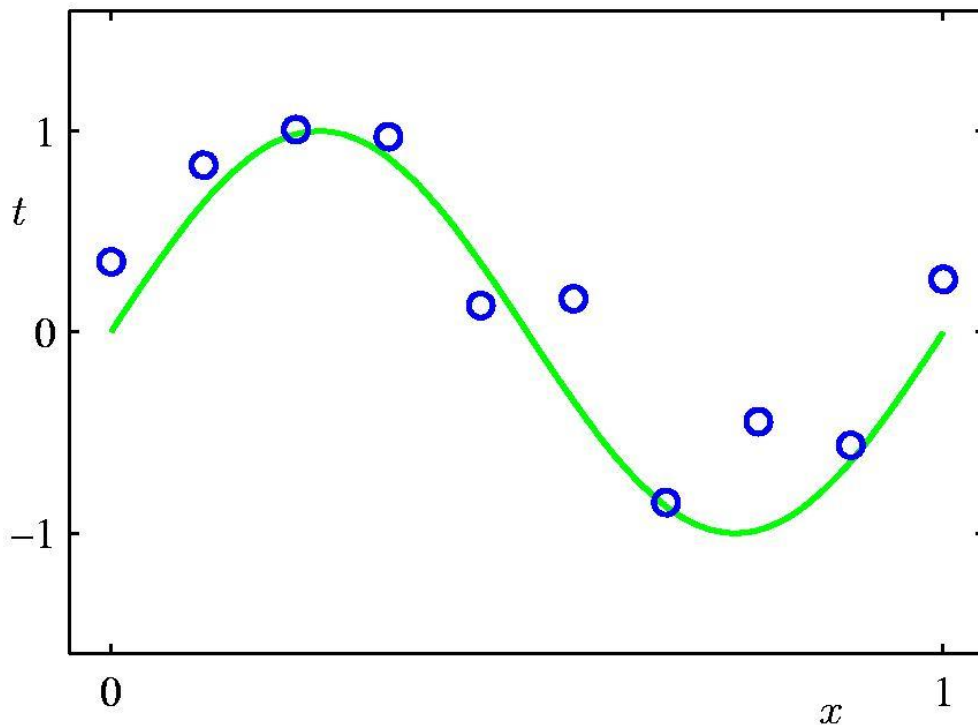
<https://echarts.apache.org/zh/index.html>

一维数据可视化 – 拟合



失业率数据的散点图与直线拟合

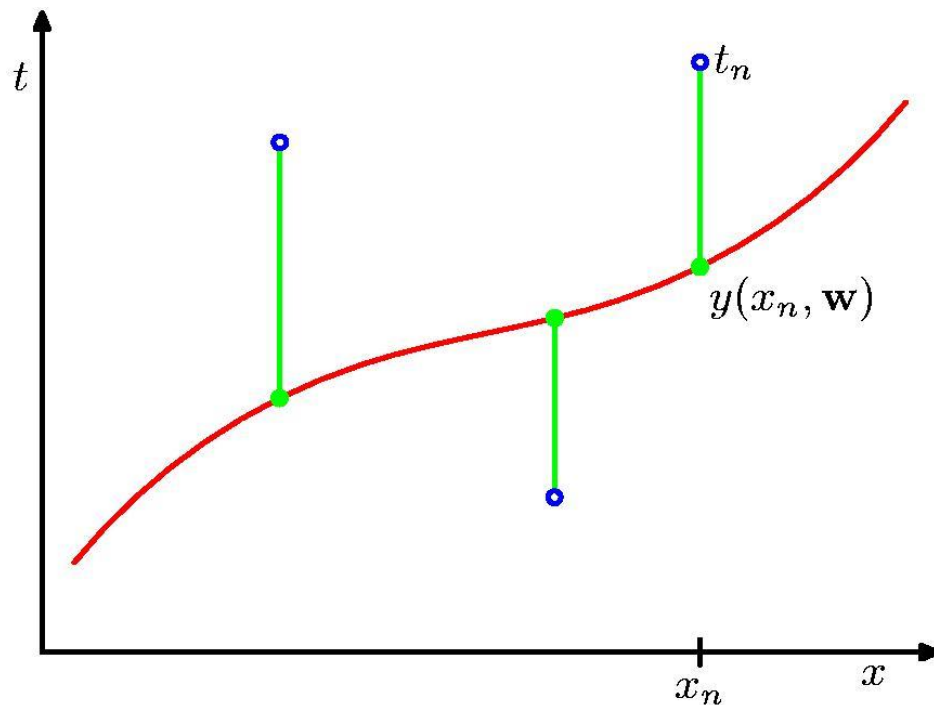
一维数据可视化 – 多项式拟合



$$y(x, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_j x^j$$

多项式函数与多项式基

一维数据可视化 – 多项式拟合

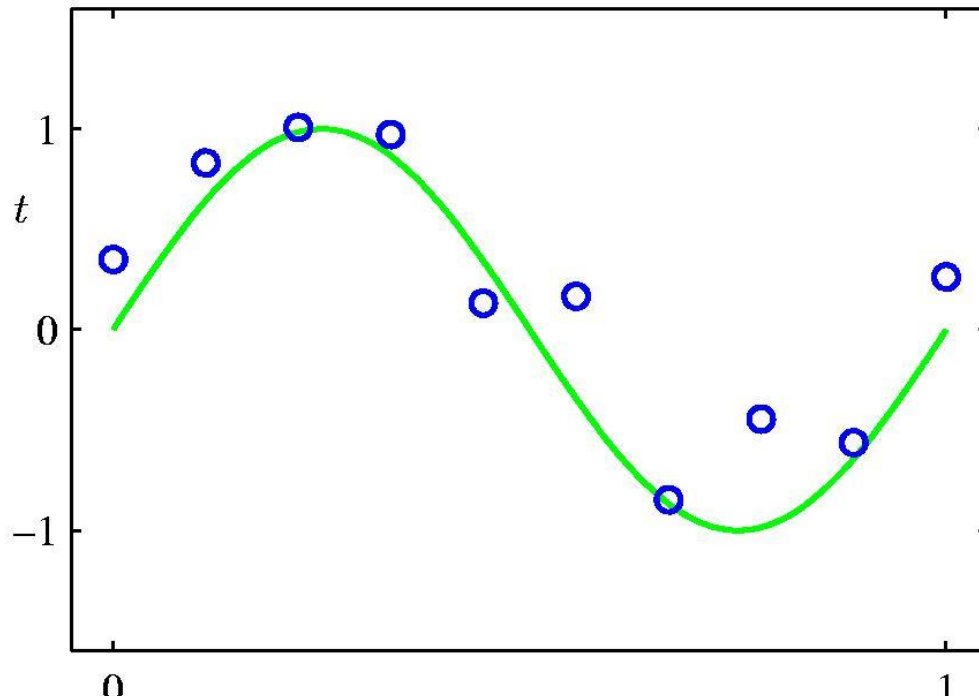


$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2$$

平方和误差函数



一维数据可视化 – 多项式拟合



$$\mathbf{p}(x) = \begin{pmatrix} x^0 \\ x^1 \\ \vdots \\ x^M \end{pmatrix}$$

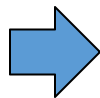
$$\mathbf{w} = \begin{pmatrix} w^0 \\ w^1 \\ \vdots \\ w^M \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y^0 \\ y^1 \\ \vdots \\ y^N \end{pmatrix}$$

$$y_j = \mathbf{p}(x_j) \cdot \mathbf{w}$$

$$y(x, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_j x^j$$

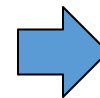
$$\mathbf{y} = \mathbf{P}(\mathbf{x}) \cdot \mathbf{w}$$

Matrix form



$$\arg \min_{\mathbf{w}} \|\mathbf{y} - \mathbf{P}(\mathbf{x}) \cdot \mathbf{w}\|^2$$

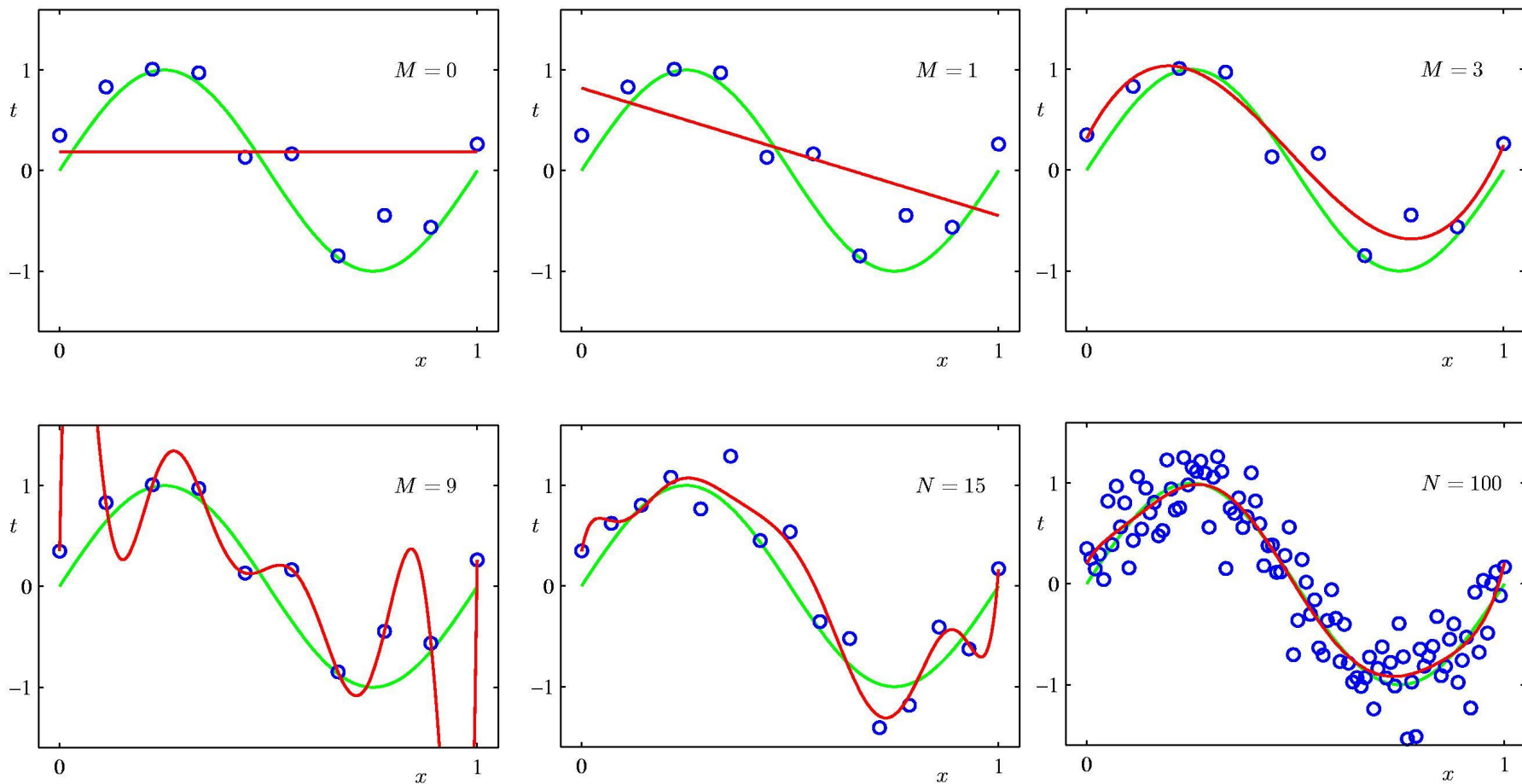
Least squares



$$\mathbf{w} = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y}$$

Normal equation

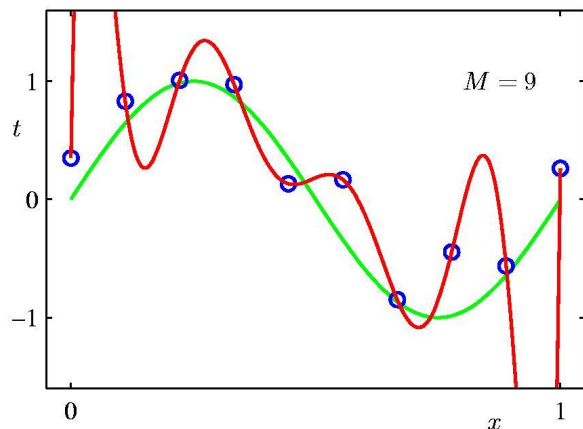
一维数据可视化 – 多项式拟合



$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2$$

平方和误差函数

一维数据可视化 – 过拟合现象

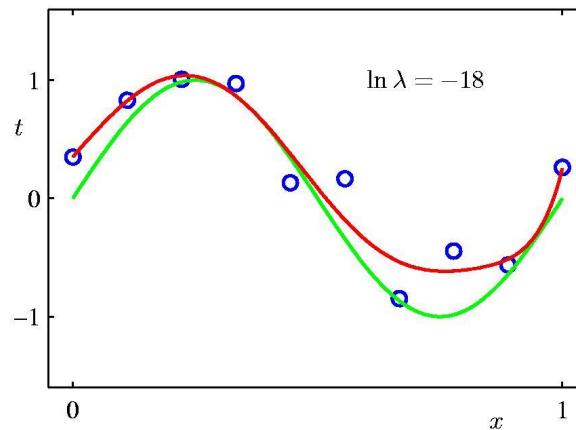
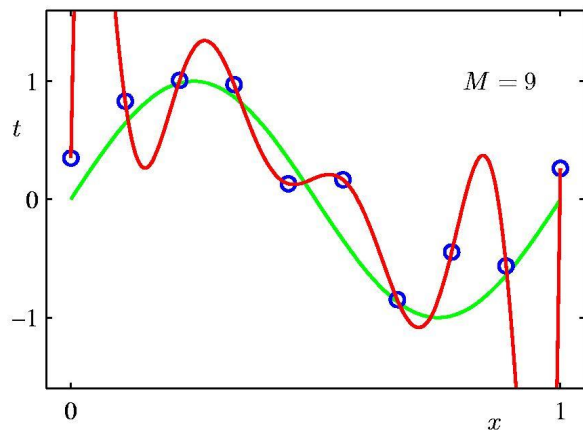


$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2$$

平方和误差函数

	$M = 0$	$M = 1$	$M = 3$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

一维数据可视化 – 过拟合现象及其解法



$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2 \quad \tilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

平方和误差函数

正则函数校正

一维数据可视化 – LOESS拟合



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Journal of the American Statistical Association
Volume 74, 1979 - Issue 368

1,494 Views
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Theory and Method
Robust Locally Weighted Regression and Smoothing Scatterplots
William S. Cleveland
Pages 829-836 | Received 01 Mar 1978, Published online: 05 Apr 2012
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Abstract

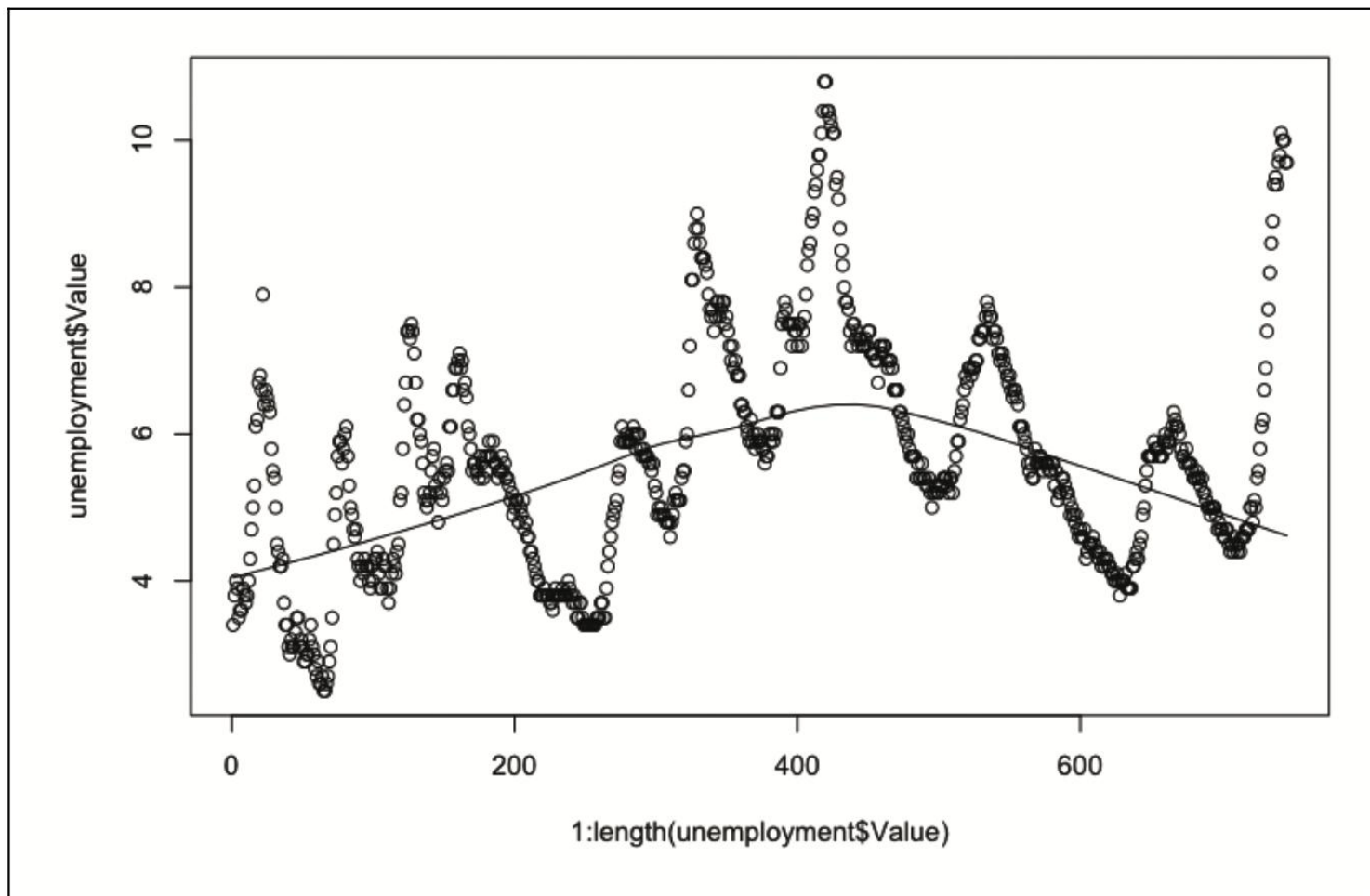
The visual information on a scatterplot can be greatly enhanced, with little additional cost, by computing and plotting smoothed points. Robust locally weighted regression is a method for smoothing a scatterplot, (x_i, y_i) , $i = 1, \dots, n$, in which the fitted value at z_k is the value of a polynomial fit to the data using weighted least squares, where the weight for (x_i, y_i) is large if x_i is close to x_k and small if it is not. A robust fitting procedure is used that guards against deviant points distorting the smoothed points. Visual, computational, and statistical issues of robust locally weighted regression are discussed. Several examples, including data on lead intoxication, are used to illustrate the methodology.

Key Words: Graphics, Scatterplots, Nonparametric regression, Smoothing, Robust estimation

William Cleveland

Robust Locally Weighted Regression and Smoothing Scatterplots
Journal of the American statistical association 74.368 (1979): 829-836.

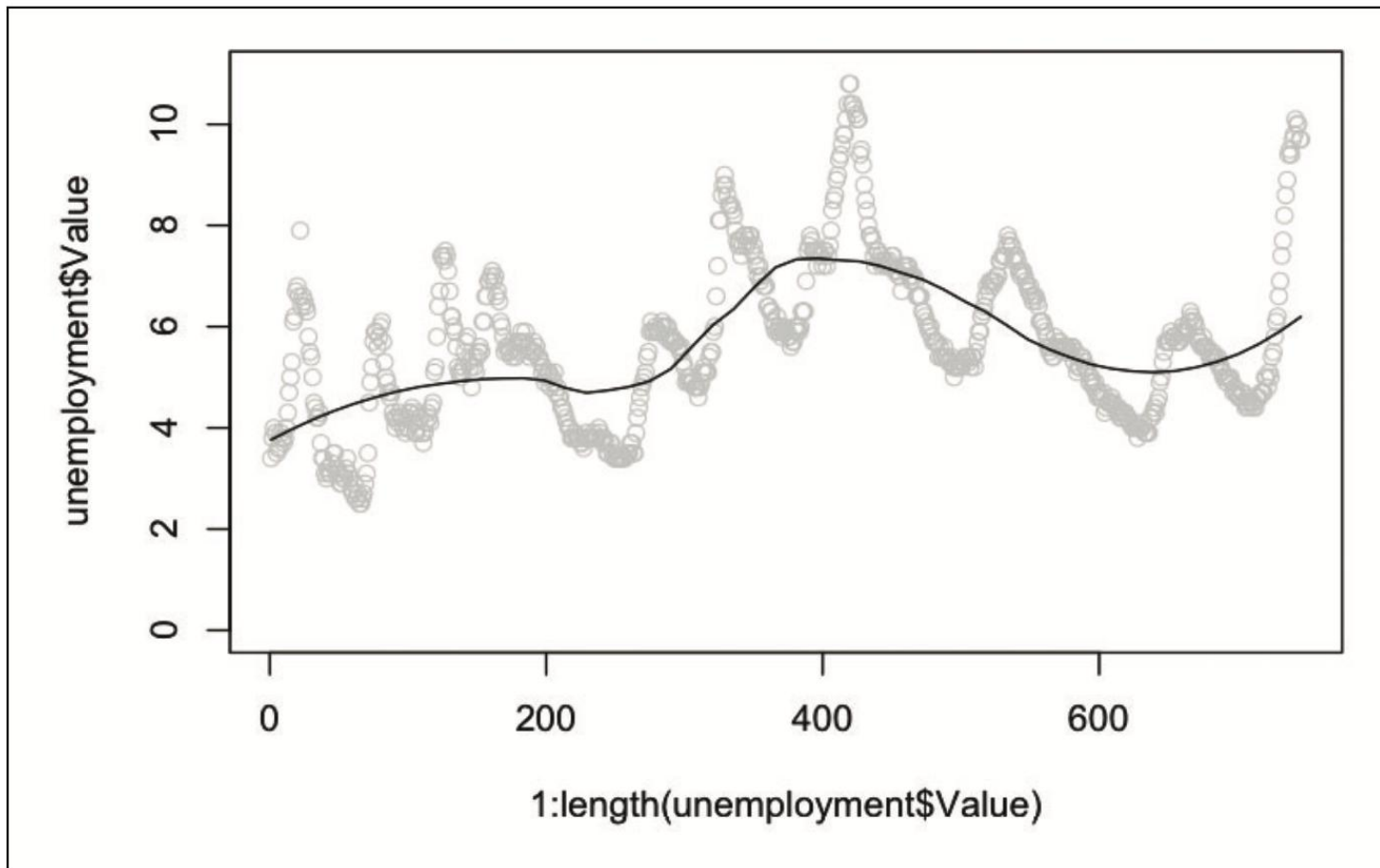
一维数据可视化 – LOESS拟合例



R语言实现:

```
scatter.smooth(x=1:length(data$Value), y=data$Value)
```

一维数据可视化 – LOESS拟合例

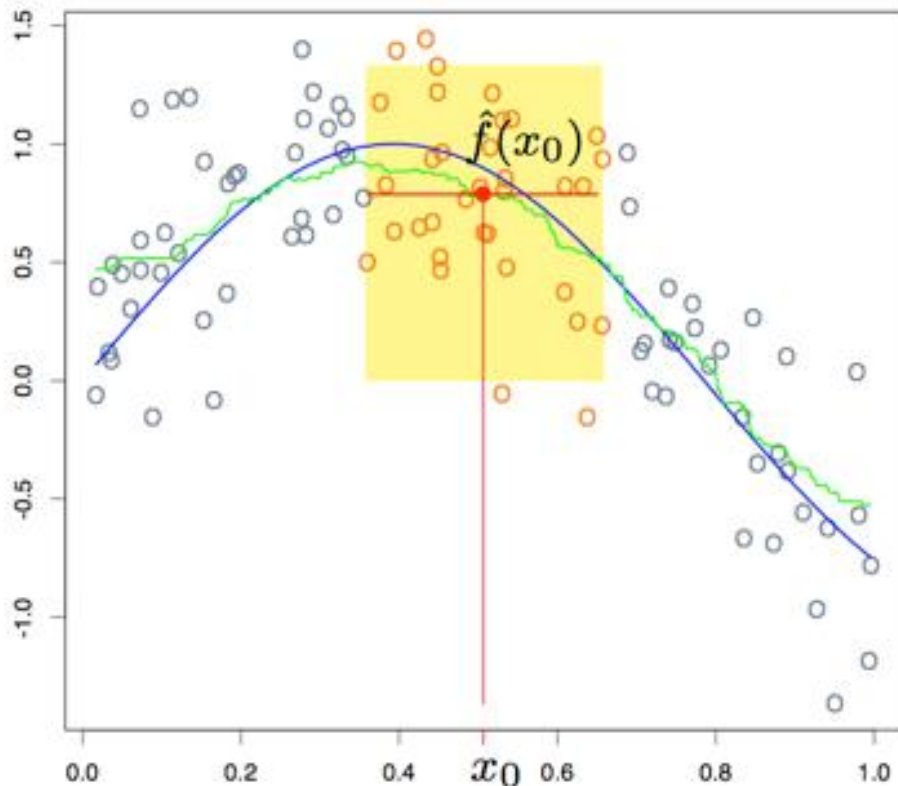


R语言实现（更多参数调整）：

```
scatter.smooth(x=1:length(data$Value), y=data$Value,  
degree=2, span=0.5)
```

一维数据可视化 – LOESS拟合

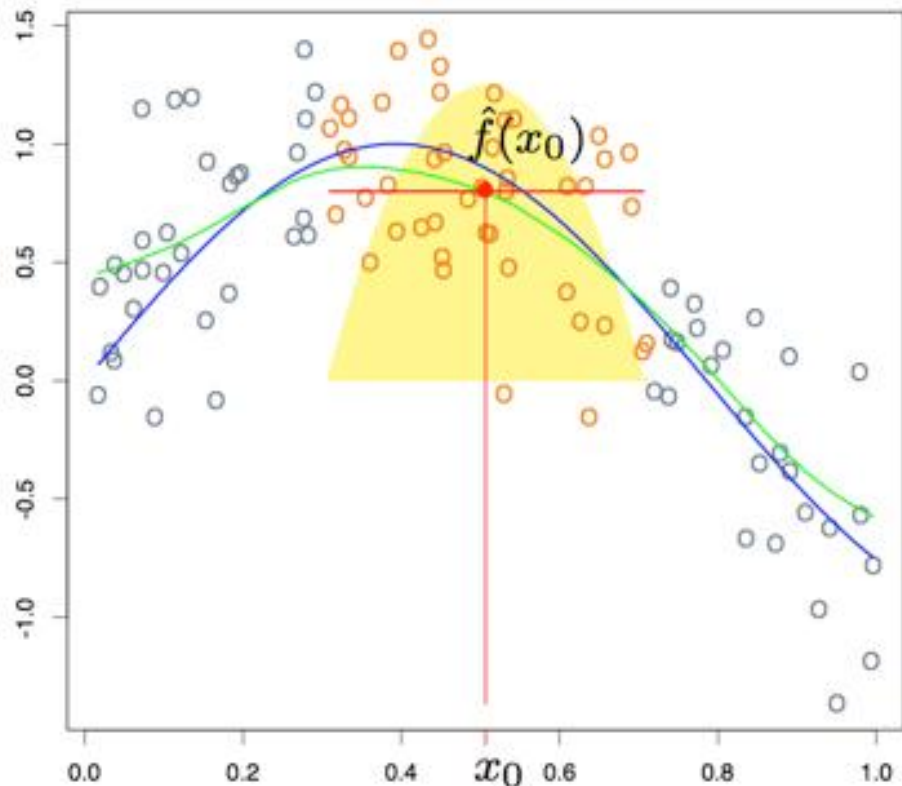
Nearest-Neighbor Kernel



K-NN平均回归造成结果不连续

$$\hat{f}(x) = Ave(y_i | x_i \in N_k(x))$$

Epanechnikov Kernel



Kernel方法带来了连续光滑的结果

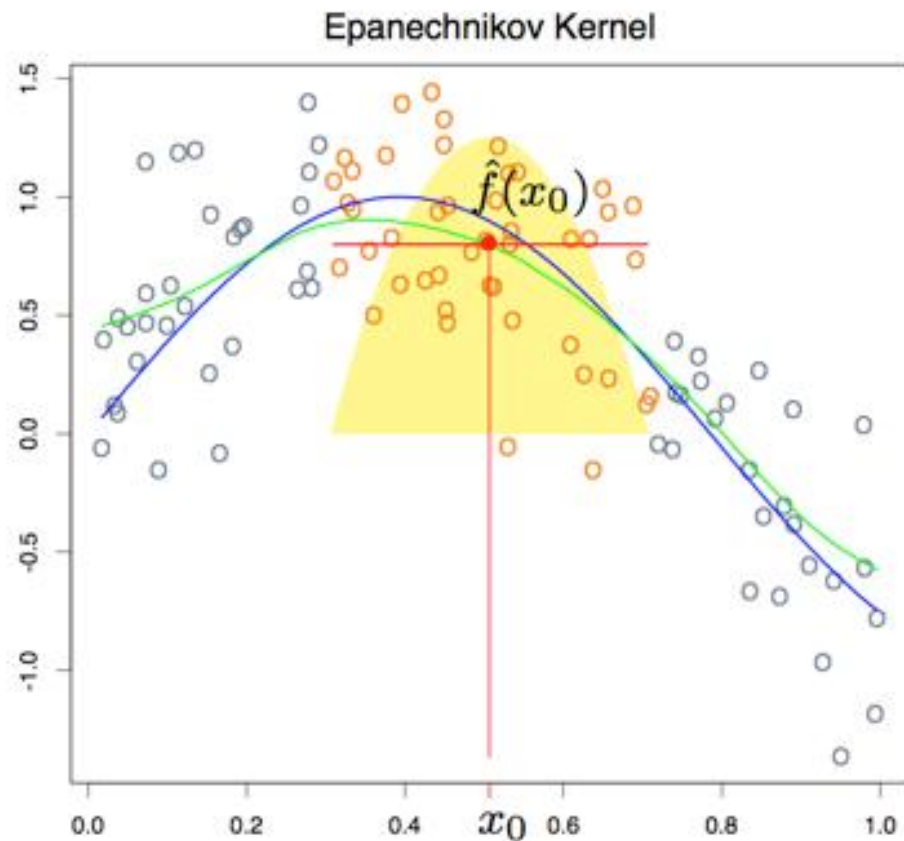
$$\hat{f}(x_0) = \frac{\sum_{i=1}^N K_\lambda(x_0, x_i) y_i}{\sum_{i=1}^N K_\lambda(x_0, x_i)}$$

一维数据可视化 – LOESS拟合

$$\hat{f}(x_0) = \frac{\sum_{i=1}^N K_{\lambda}(x_0, x_i) y_i}{\sum_{i=1}^N K_{\lambda}(x_0, x_i)}$$

$$K_{\lambda}(x_0, x_i) = D\left(\frac{|x_0 - x_i|}{\lambda}\right)$$
$$D(t) = \begin{cases} \frac{3}{4}(1 - t^2) & \text{for } |t| < 1 \\ 0 & \text{otherwise} \end{cases}$$

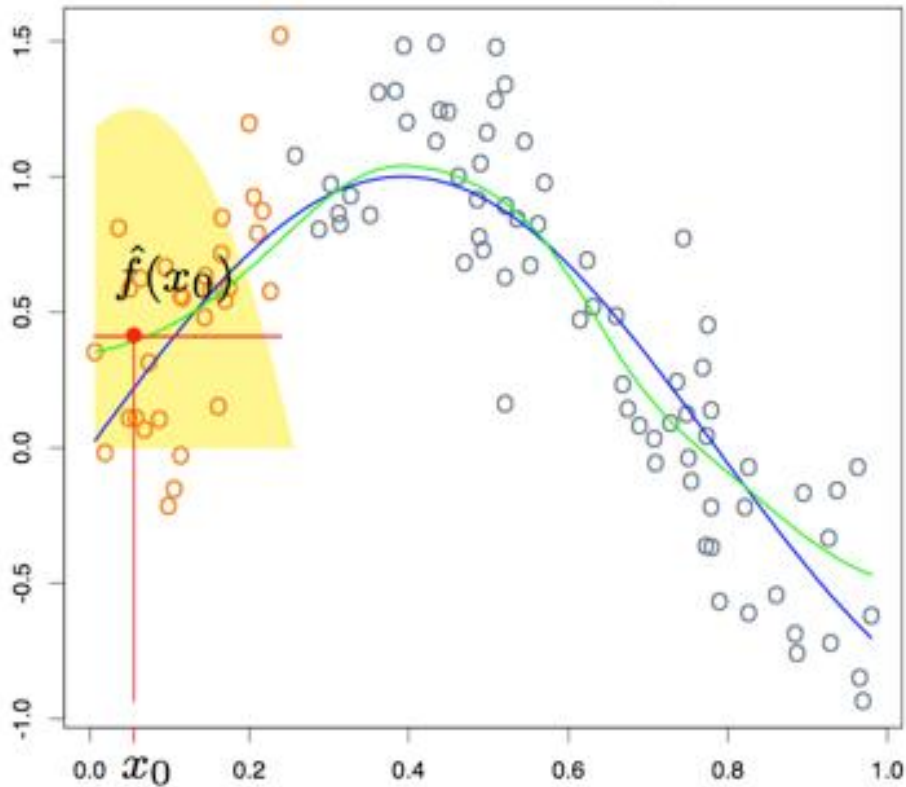
Epanechnikov 二次kernel函数



kernel方法带来了连续光滑的结果

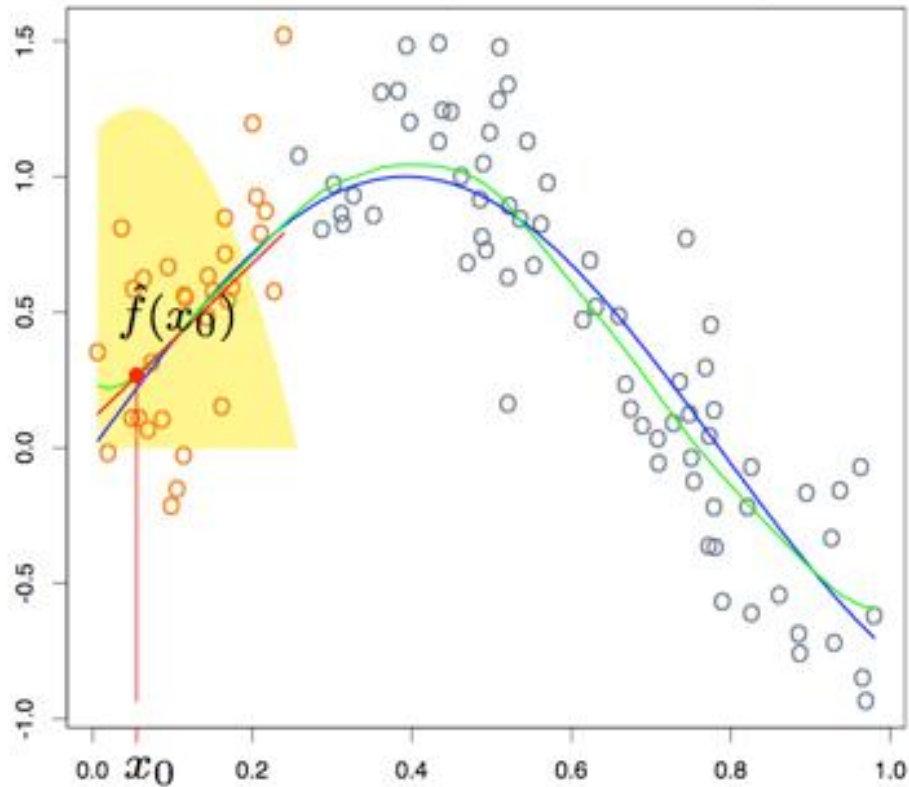
一维数据可视化 – LOESS拟合

N-W Kernel at Boundary



Kernel方法存在边界问题
开始与结束区段的点
其左右邻域是不对称的
导致了平滑后的值偏大或偏小

Local Linear Regression at Boundary



需要对权值做再修正

一维数据可视化 – LOESS拟合

$$\min_{\beta} \sum_{i=1}^N K_{\lambda}(x_0, x_i) [y_i - \sum_{j=0}^d \beta_j x_i^j]^2$$

$$B = \begin{pmatrix} 1 & x_1 & \cdots & x_1^d \\ 1 & x_2 & \cdots & x_2^d \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_N & \cdots & x_N^d \end{pmatrix}$$

$$W_{x_0} = \begin{pmatrix} K_{\lambda}(x_0, x_1) & 0 & \cdots & 0 \\ 0 & K_{\lambda}(x_0, x_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & K_{\lambda}(x_0, x_N) \end{pmatrix}$$

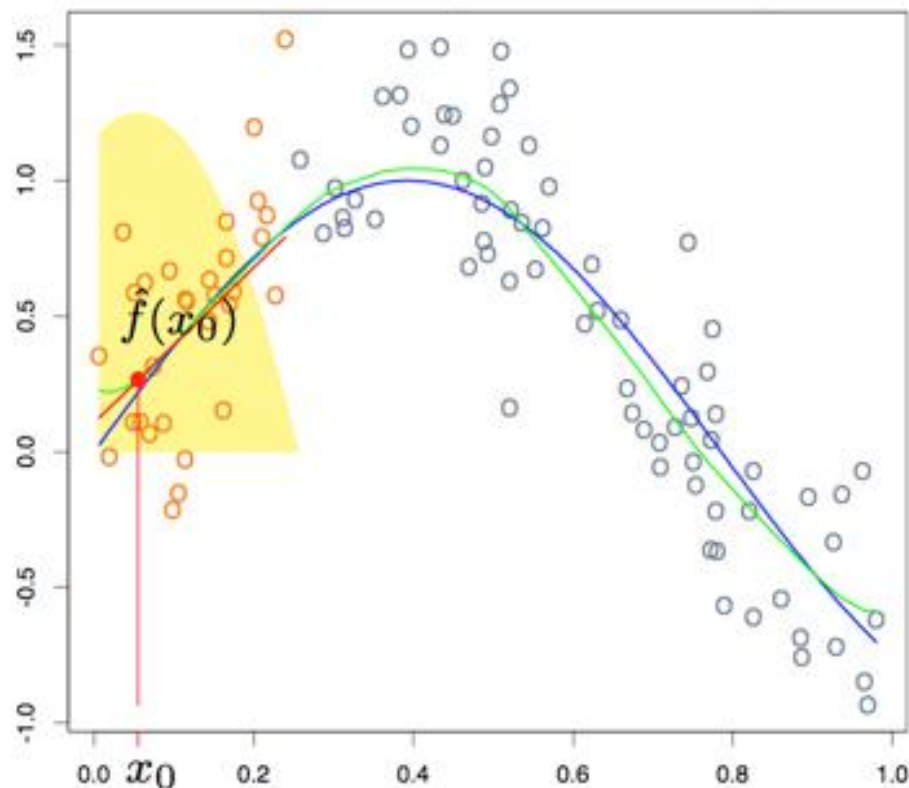
$$\Delta = (\beta_0, \beta_1, \dots, \beta_d)^T$$

$$Y = (y_1, y_2, \dots, y_N)^T$$



$$\min_{\Delta} (Y - B\Delta)^T W_{x_0} (Y - B\Delta)$$

Local Linear Regression at Boundary



需要对权值做再修正

$$\hat{f}(x_0) = \sum_{j=0}^d \beta_j x_0^j$$

一维数据可视化 – LOESS拟合

$$\Delta = (B^T W_{x_0} B)^{-1} (B^T W_{x_0} Y)$$

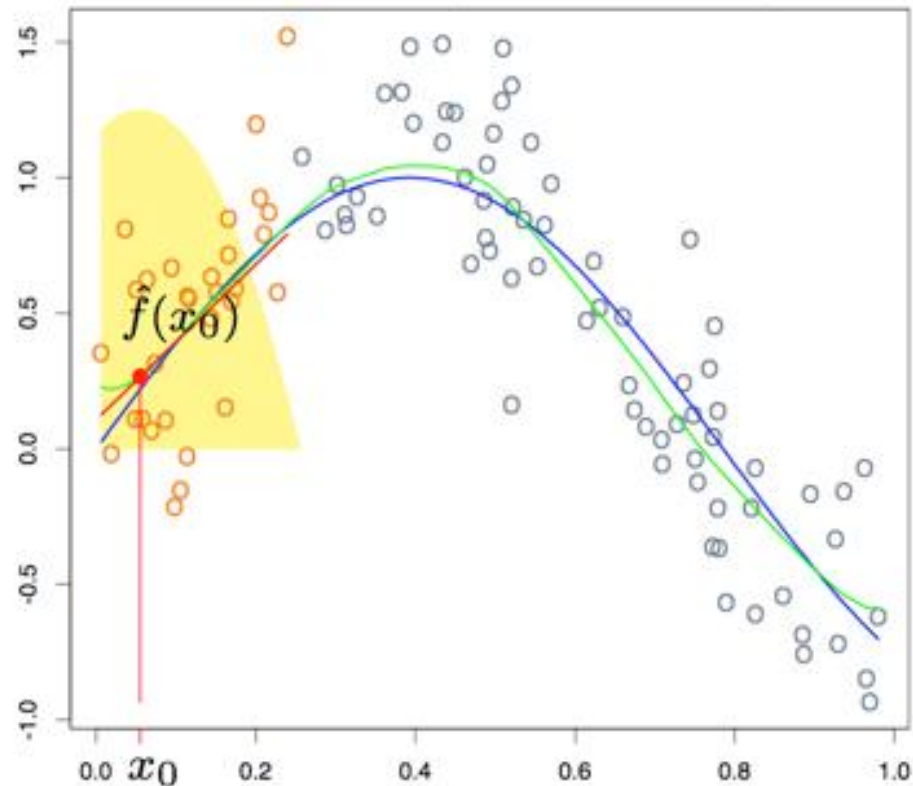
$$e(x_0) = (1, x_0, \dots, x_0^d)$$

$$\begin{aligned} \hat{f}(x_0) &= e(x_0) (B^T W_{x_0} B)^{-1} (B^T W_{x_0} Y) \\ &= \sum_i w_i(x_0) y_i \end{aligned}$$



$$\min_{\Delta} (Y - B\Delta)^T W_{x_0} (Y - B\Delta)$$

Local Linear Regression at Boundary



需要对权值做再修正

$$\hat{f}(x_0) = \sum_{j=0}^d \beta_j x_0^j$$

一维数据可视化 – Robust LOESS拟合



计算残差

$$e_i = y_i - \hat{f}(x_i)$$

根据残差计算Robust Weight

$$\delta_i = B(e_i/6s)$$

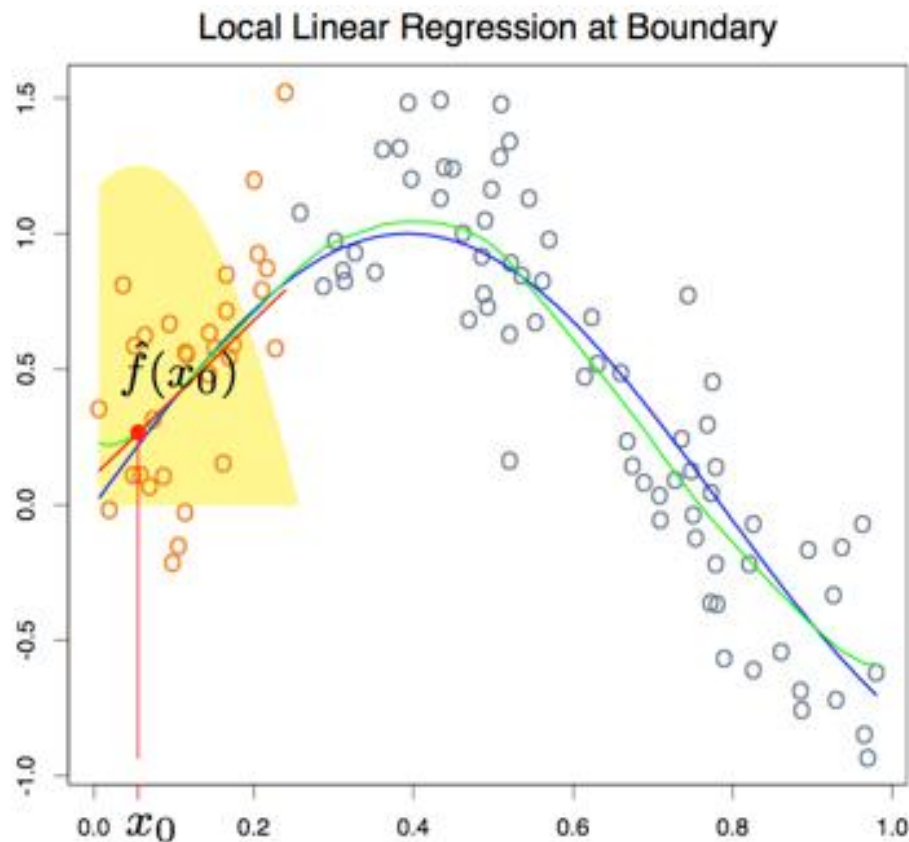
其中

s 为残差绝对值序列 $|e_i|$ 的中位数

B 函数为bisquare函数

$$B(u) = \begin{cases} (1 - u^2)^2 & \text{for } 0 \leq u < 1 \\ 0 & \text{for } u \geq 1 \end{cases}$$

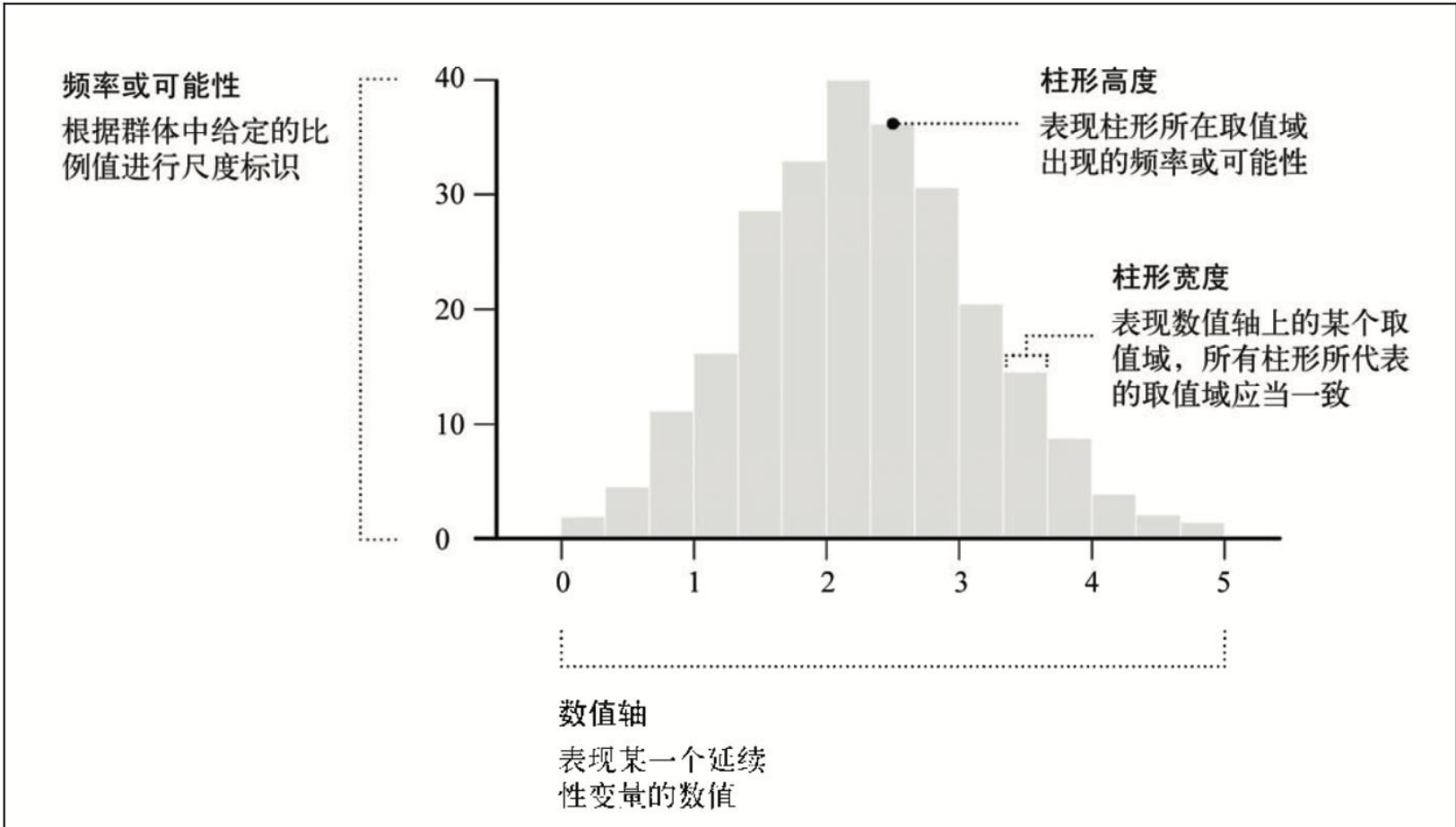
用robustness weight乘以
kernel weight作为新的weight



剔除残差较大的异常点
对于回归的影响



一维数据可视化 – 直方图



直方图的基本框架

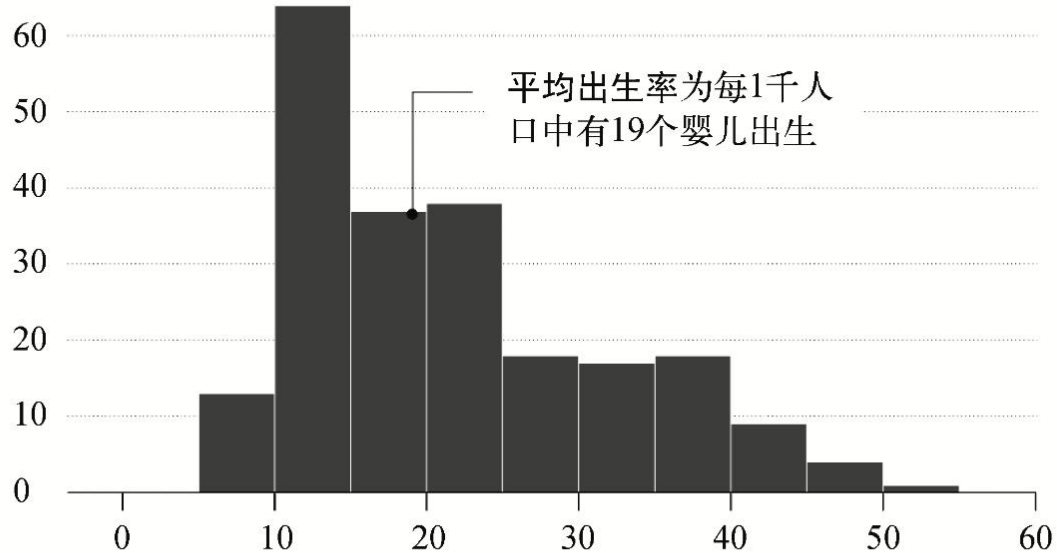


一维数据可视化 – 直方图例

全球出生率分布图

2008年，绝大多数国家每1千人口中新出生婴儿的数量都少于25。但仍然还有很多发展中国家的女性倾向于要更多的孩子。

国家数量

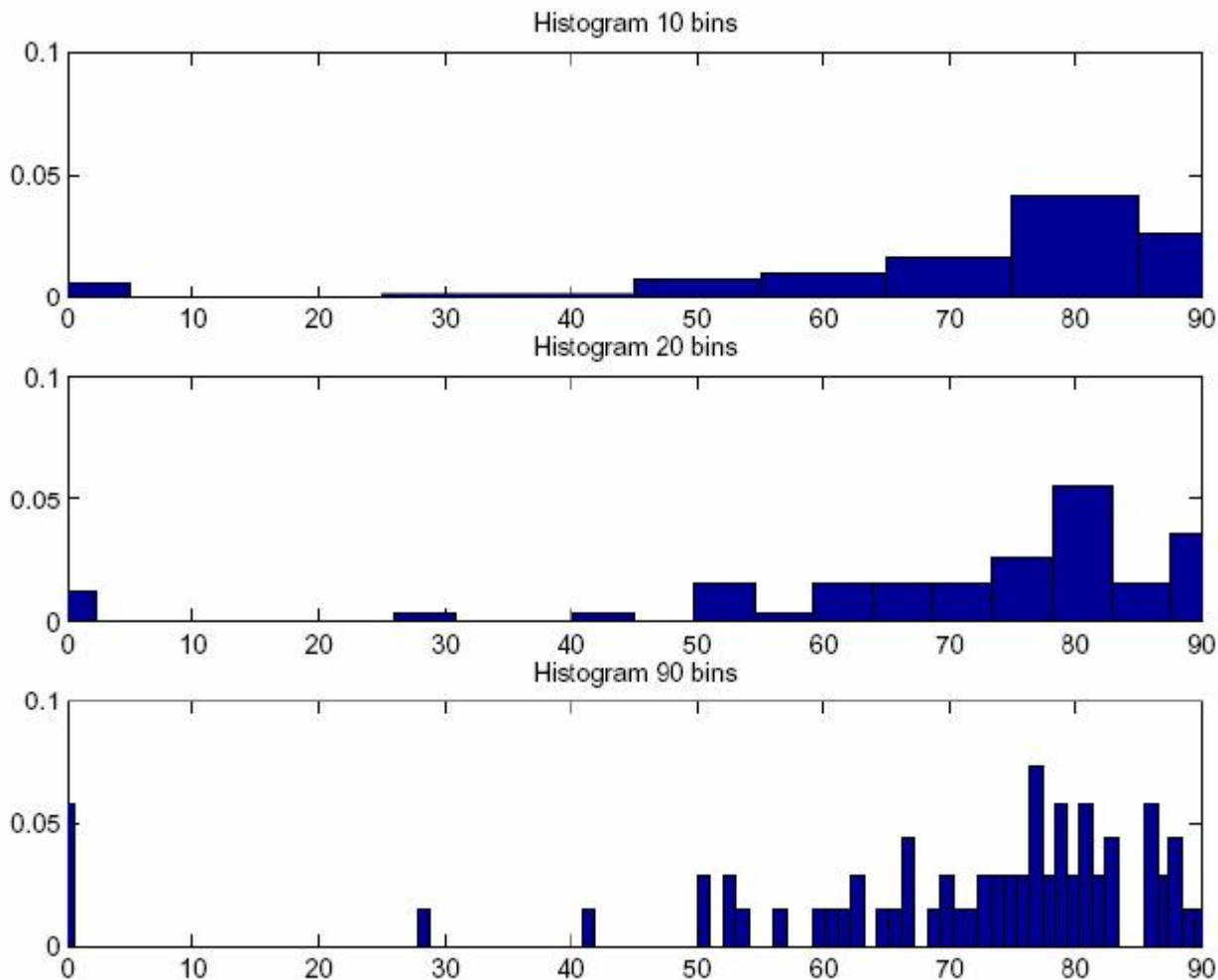


Source: The World Bank

每1千人口中新出生婴儿的数量

全球出生率分布图

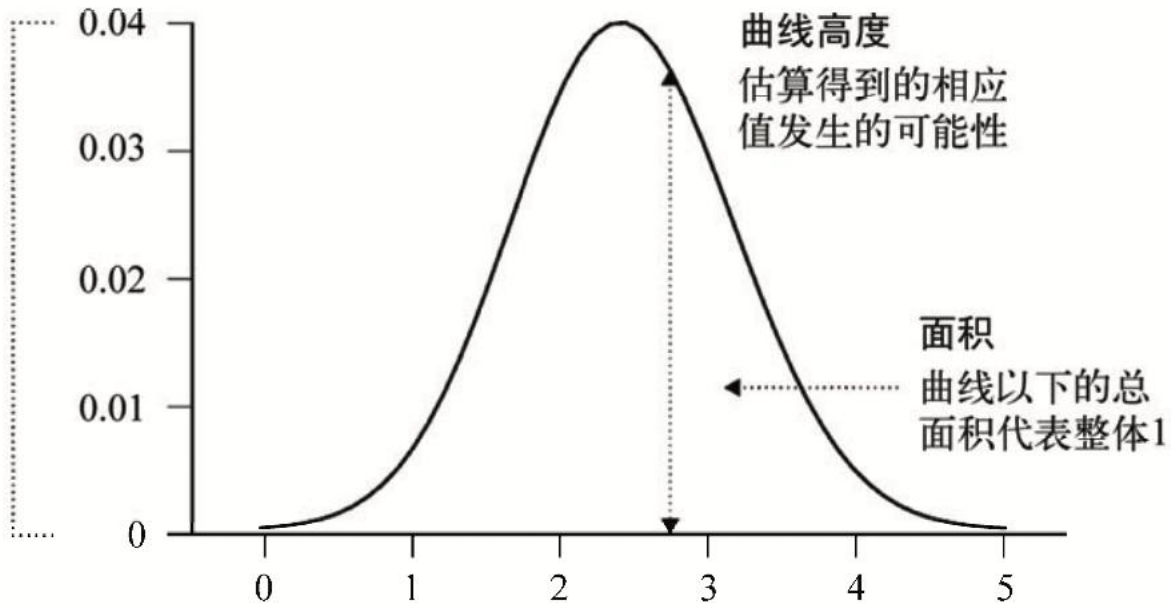
一维数据可视化 – 直方图的缺点



不同采样率下的直方图

一维数据可视化 – 密度图

可能性大小
群体中给定的
比例值

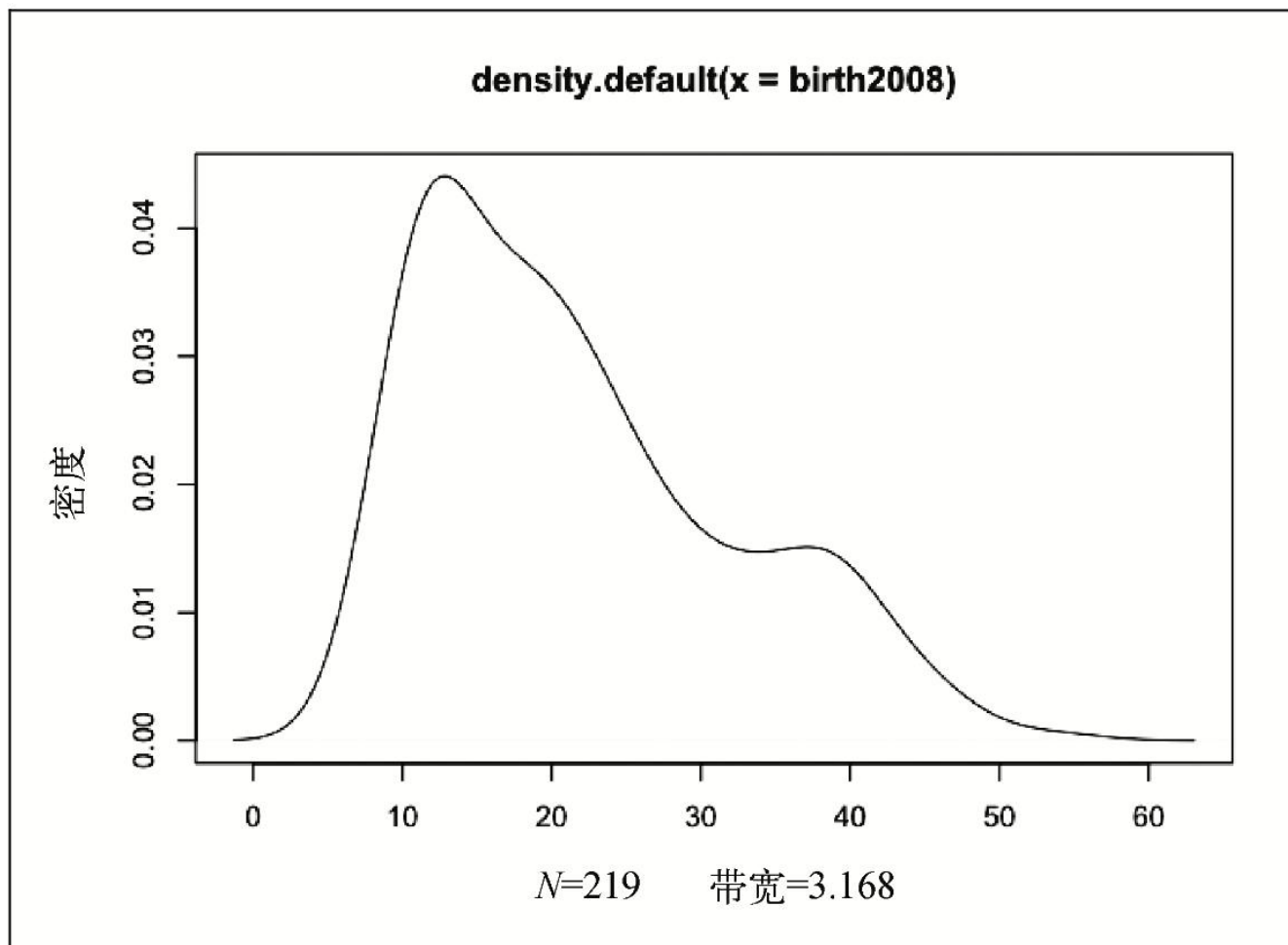


数值轴

表现某一个延续性变量的数值

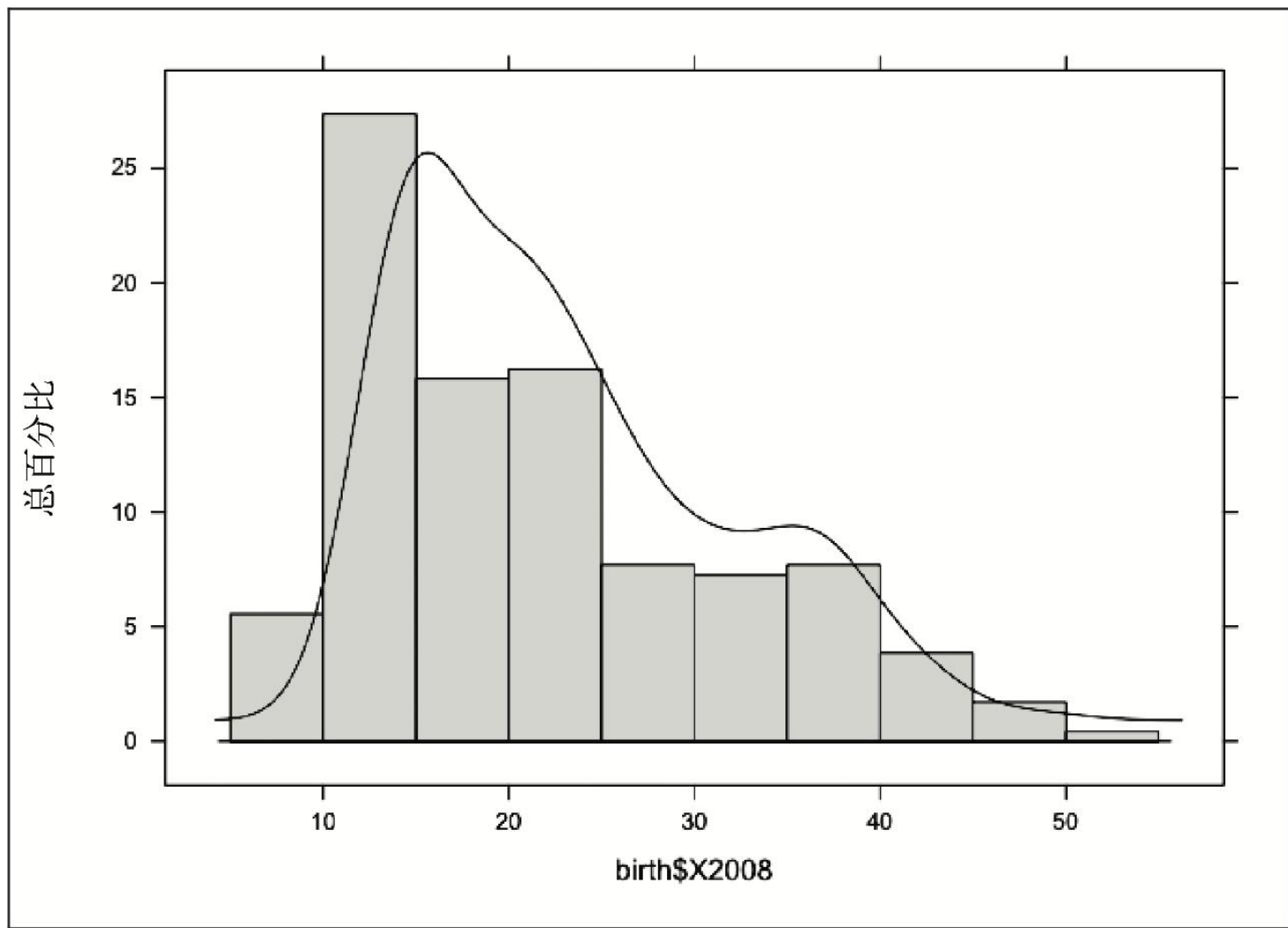
密度图的基本框架

一维数据可视化 – 密度图例



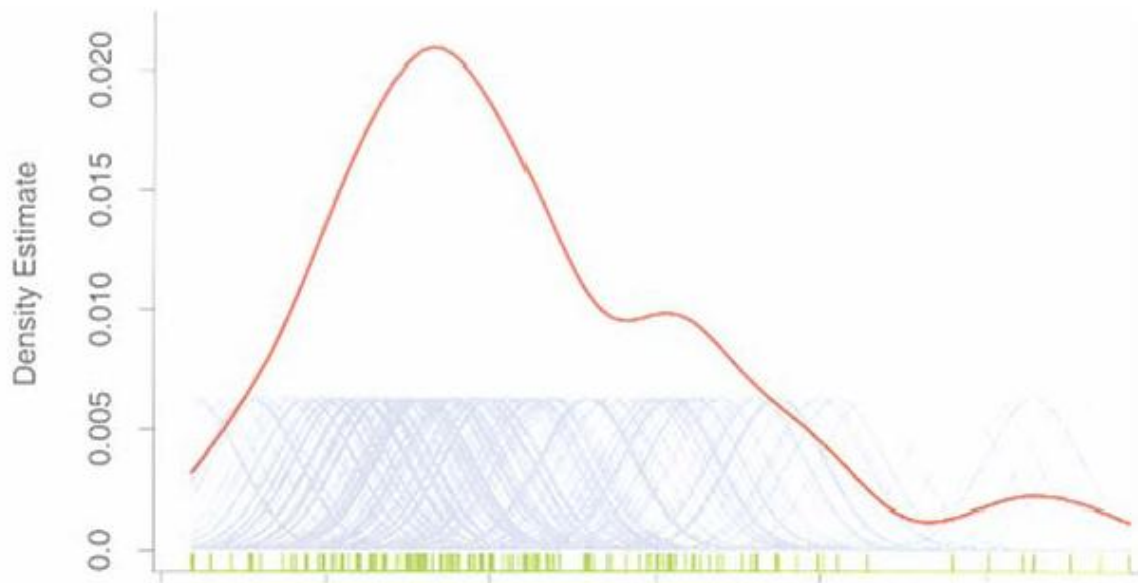
全球出生率的密度图

一维数据可视化 – 图例



全球出生率的直方图与密度图混合

Parzen example



from Hastie *et al.*

密度图与非参数估计

■ Multivariate kernel density estimation

$$f(x) = \frac{1}{nh^d} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - x_i}{h}\right)$$

■ Kernels

■ Gaussian $K_N = (2\pi)^{-d/2} \exp(-\frac{1}{2}\|\mathbf{x}\|^2)$

■ Epanechnikov

$$K_E = \begin{cases} 1/2c_d^{-1}(d+2)(1-\|\mathbf{x}\|^2) & \text{if } \|\mathbf{x}\| < 1 \\ 0 & \text{otherwise} \end{cases}$$



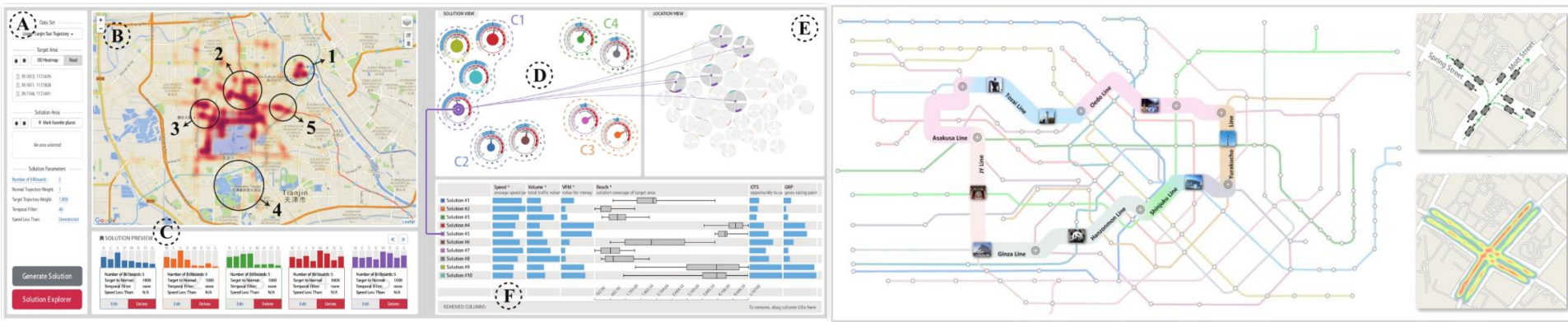
主要参考资料 《大数据可视化建模》

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谢谢

Thank You

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开源：

<https://github.com/hongxin/vizmodeling>